Lecture 06:

Concentration Inequalities and Generalization

Matthew Wicker

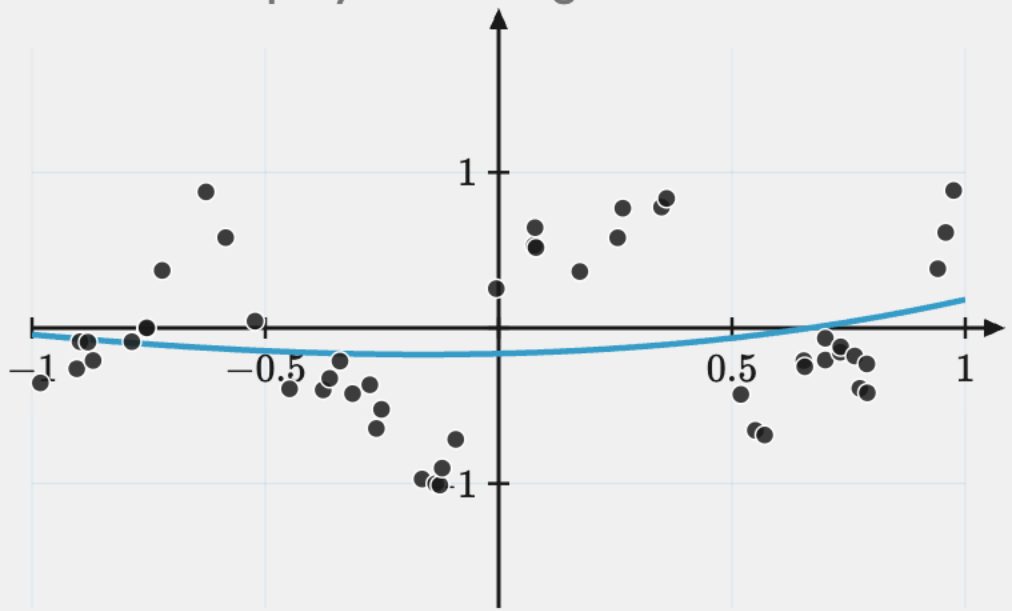
22 October 2026

Logistics

- Office hours are slightly changed today and will be from 3pm to 4:30pm in my office ACEX 258.
- The coursework will be announced and assigned on Tuesday.
- It will be a group project done in groups of 3 to 4 students.

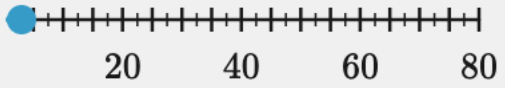
Fitting a complex regression dataset

polynomial regression

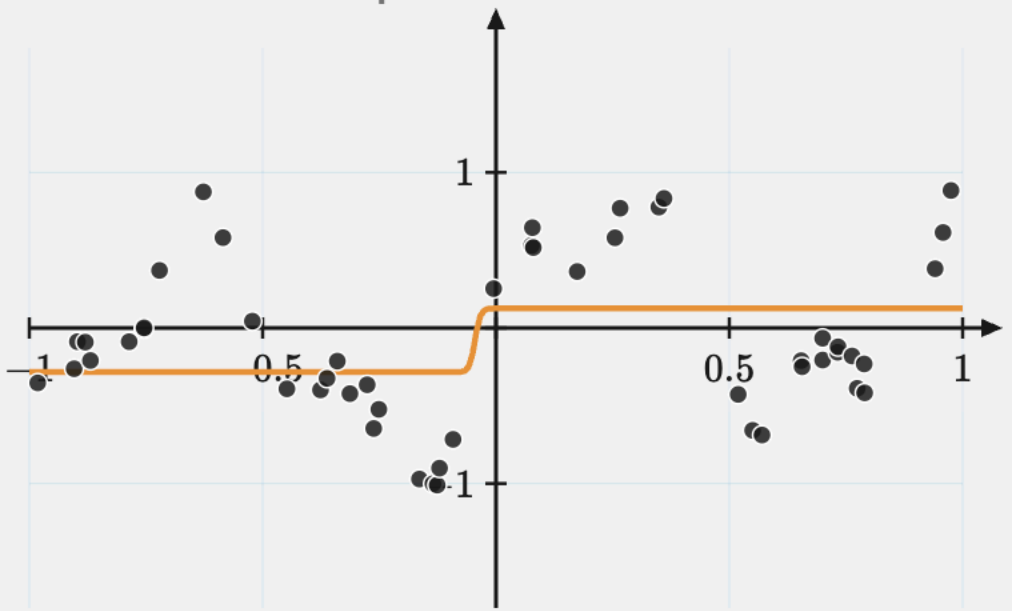


degree $k = 3$: training error 0.258

$k = 3$

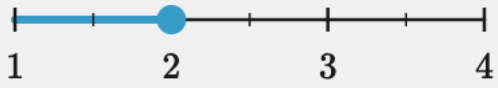


a deep neural network

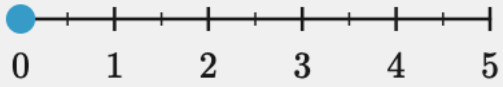


$d = 2$ hidden layers of $w = 1$ units: training error 0.224

$d = 2$

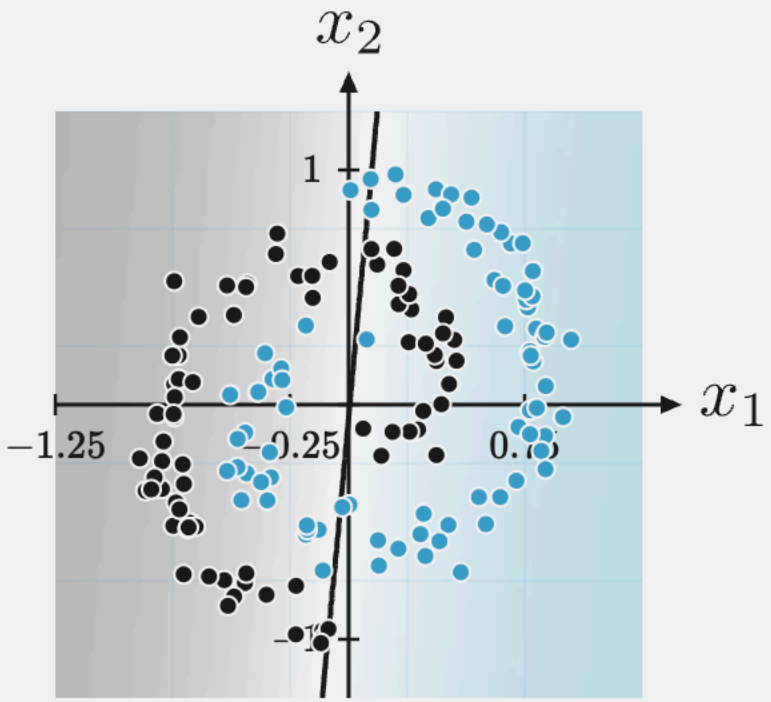


$log_2\, w = 0$



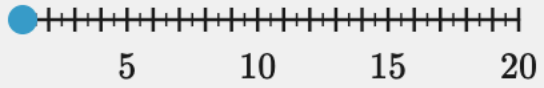
Fitting a complex classification dataset

logistic regression, polynomial features

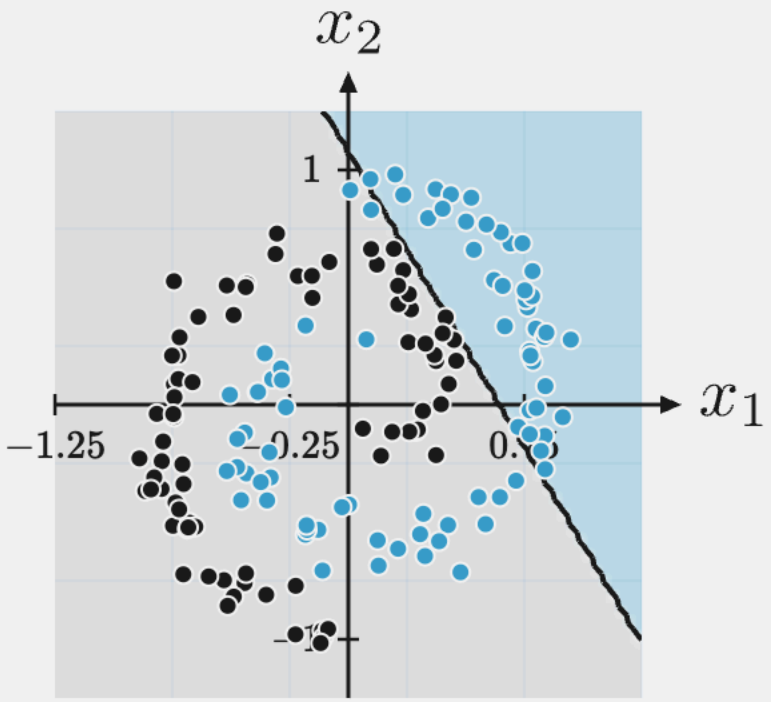


degree $k = 1$: loss 0.546, 67% right

$k = 1$



a deep neural network



$d = 2$ layers of $w = 1$: loss 0.491, 74% right

$d = 2$

$log_2\, w = 0$



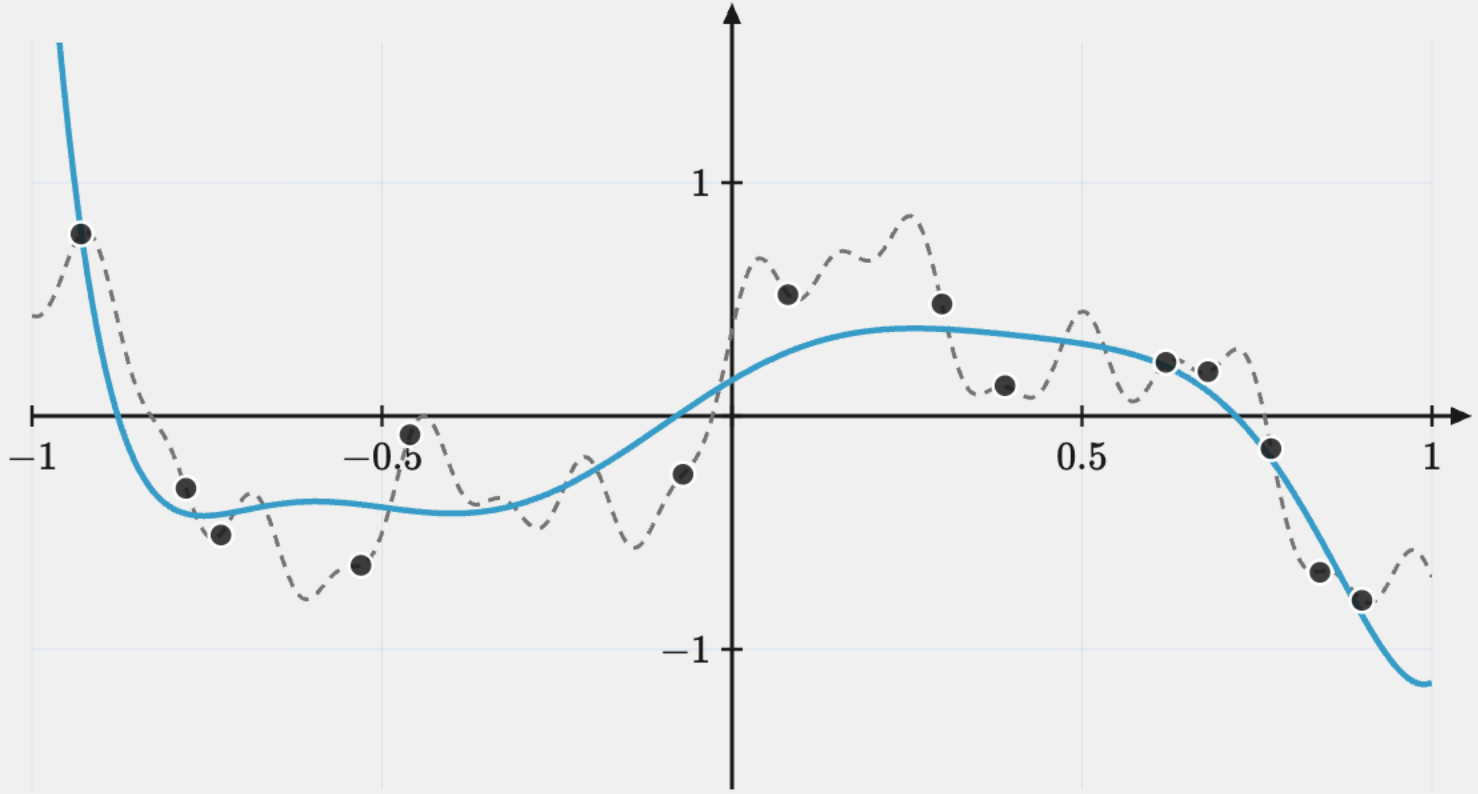
Universal function approximation

Theorem 6.1.1 (Weierstrass approximation theorem). Let f be a continuous function on a closed interval $[a, b]$. For any $\varepsilon > 0$, there exists a polynomial function $P(x)$ such that

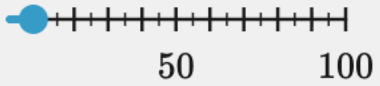
$$\|f - P\|_\infty = \sup_{x \in [a,b]} |f(x) - P(x)| < \varepsilon.$$

In other words, the polynomial $P(x)$ can uniformly approximate f on the interval $[a, b]$.

Polynomials can fit anything



$k = 8$



what to fit:

f, a chaotic function

P, a polynomial of degree k

degree $k = 8$: training error 0.03

Universal function approximation

Theorem 6.1.1 (Weierstrass approximation theorem). Let f be a continuous function on a closed interval $[a, b]$. For any $\varepsilon > 0$, there exists a polynomial function $P(x)$ such that

$$\|f - P\|_\infty = \sup_{x \in [a,b]} |f(x) - P(x)| < \varepsilon.$$

In other words, the polynomial $P(x)$ can uniformly approximate f on the interval $[a, b]$.

We have seen that deep neural networks can be viewed as "learned" basis functions. Do these have the same property?

1-layer neural networks as function approximators

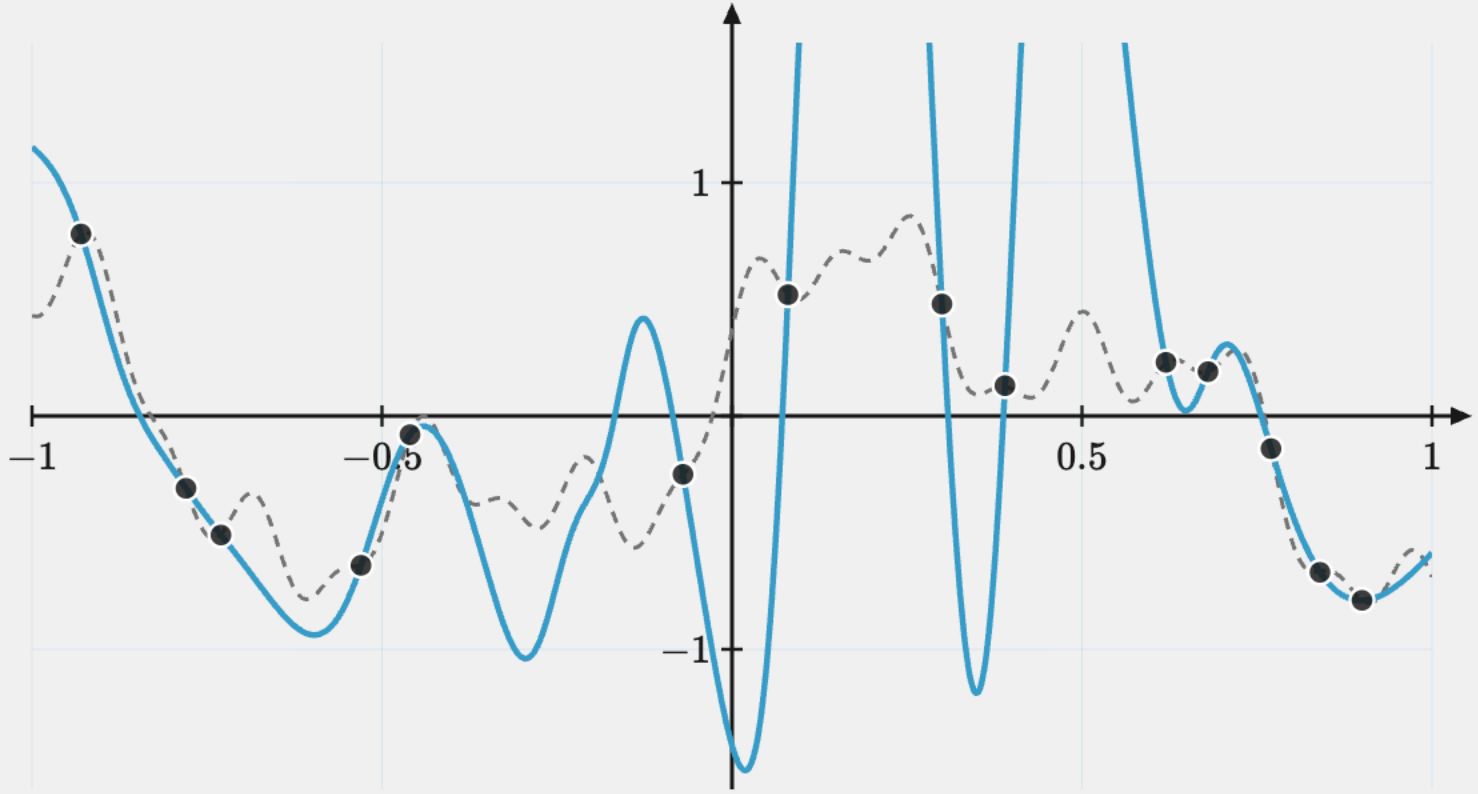
Theorem 6.1.2 (Universal approximation by sums of sigmoids; Cybenko, 1989). Let σ be a sigmoidal activation function, i.e., a function that is nonconstant, bounded, and continuously differentiable. For any continuous function f on a compact subset of $\mathbb{R}^n$, and any $\varepsilon > 0$, there exist a positive integer N, weights $v_i \in \mathbb{R}$, $\boldsymbol{w}_i \in \mathbb{R}^n$ and biases b_i such that the sum of N sigmoidal functions

$$F(\boldsymbol{x}) = \sum_{i=1}^{N} v_i \, \sigma(\boldsymbol{w}_i^\top \boldsymbol{x} + b_i)$$

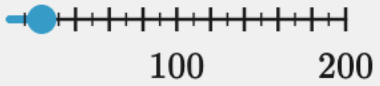
satisfies $|F(\boldsymbol{x}) - f(\boldsymbol{x})| < \varepsilon$ for all $\boldsymbol{x}$ in the compact subset.

Indeed! Deep neural networks can also approximate any function (under some mild conditions)!

Sums of sigmoids can fit anything too



$N = 20$



what to fit:

f, a chaotic function

F, a sum of N sigmoids

$N = 20$ sigmoids: training error 0.000 — through every one of the 14 points

Generalized universal function approximation

Theorem 6.1.3 (Stone–Weierstrass theorem, limited version). *Let $\mathcal{F}$ be a class of functions defined on a compact set $S \subseteq \mathbb{R}^d$. If $\mathcal{F}$ satisfies:*

1. *Each $f \in \mathcal{F}$ is continuous.*
2. *For every x, there exists $f \in \mathcal{F}$ such that $f(x) \neq 0$.*
3. *For every x, x_1 with $x \neq x_1$, there exists $f \in \mathcal{F}$ such that $f(x) \neq f(x_1)$ ($\mathcal{F}$ separates points).*
4. *$\mathcal{F}$ is closed under multiplication ($\forall f, g \in \mathcal{F}$, $\exists h \in \mathcal{F}$ such that $h(x) = f(x)g(x)$) and vector space operations ($\mathcal{F}$ is an algebra).*

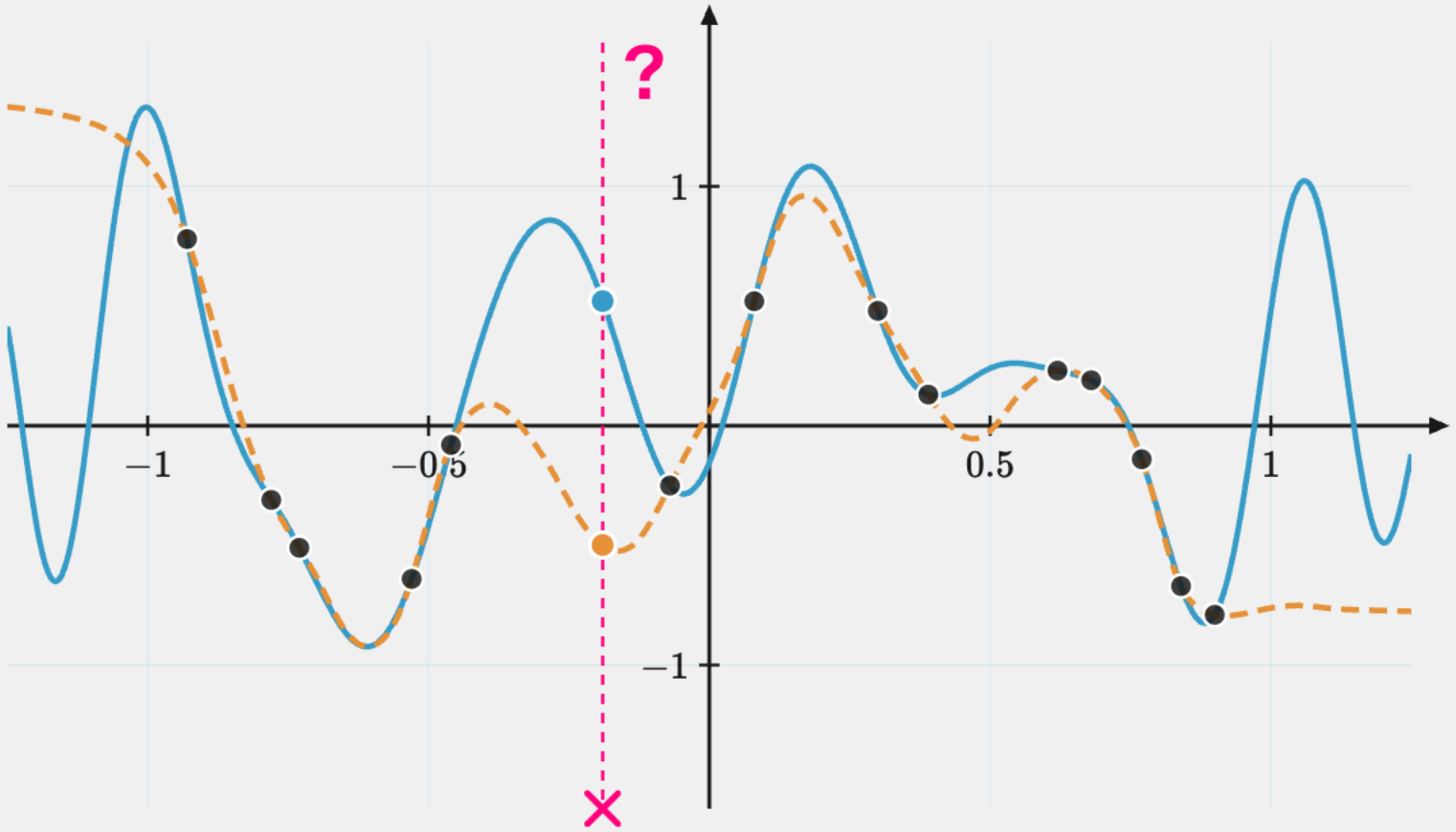
Then, for every continuous function $g : S \to \mathbb{R}$ and any $\varepsilon > 0$, there exists $f \in \mathcal{F}$ such that $\|f - g\|_\infty \leq \varepsilon$. In other words, $\mathcal{F}$ is a universal approximator.

No free lunch!

While it has a formal statement, the “no free lunch” theorem in machine learning says that any two learning algorithms ‘A’ and ‘B’ have the same loss when averaged over all possible target functions.

Put even more simply: without any knowledge of the underlying function, there is no *a priori* reason to pick a learning algorithm ‘A’ over ‘B’.

No free lunch: one unseen point



A: a polynomial of degree 30 through every point

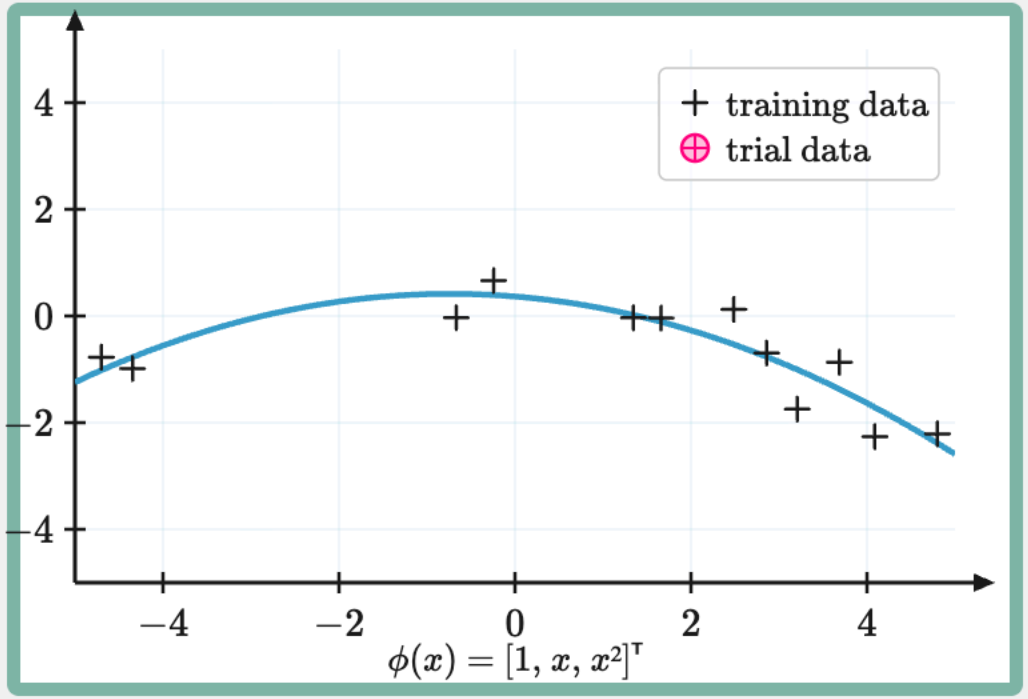
B: a sum of 100 sigmoids through every point

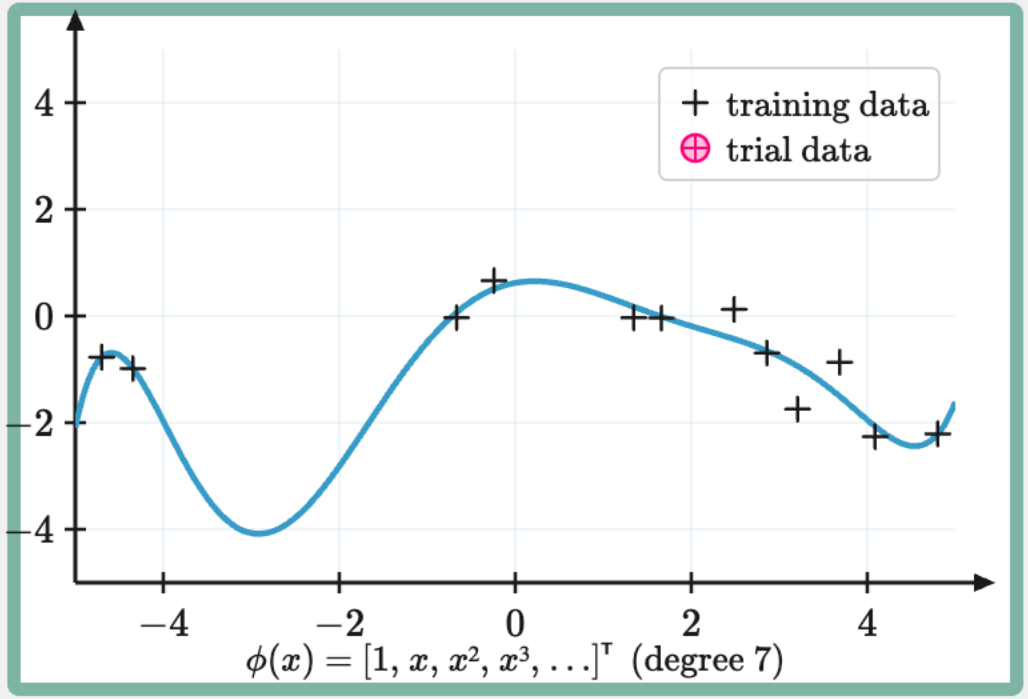
at the unseen point $x^* = -0.19$:

A says $+0.52$, B says -0.5

both fit all 14 training points (largest error < 0.001)

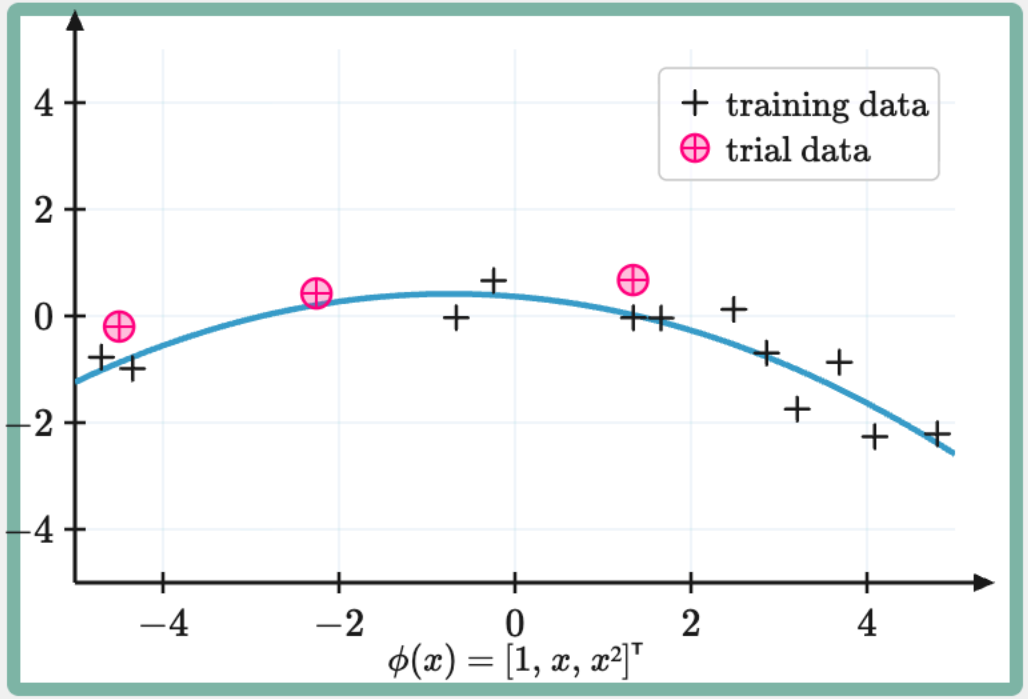
Simple thought experiment

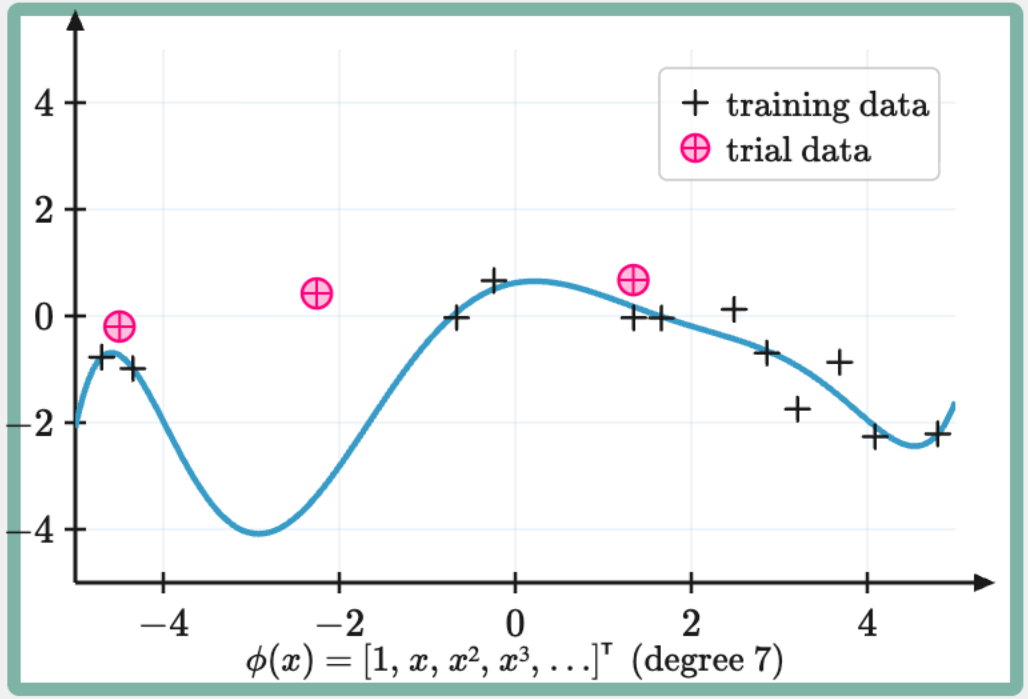




To decide which function is best, let's use both in practice and measure which one performs best

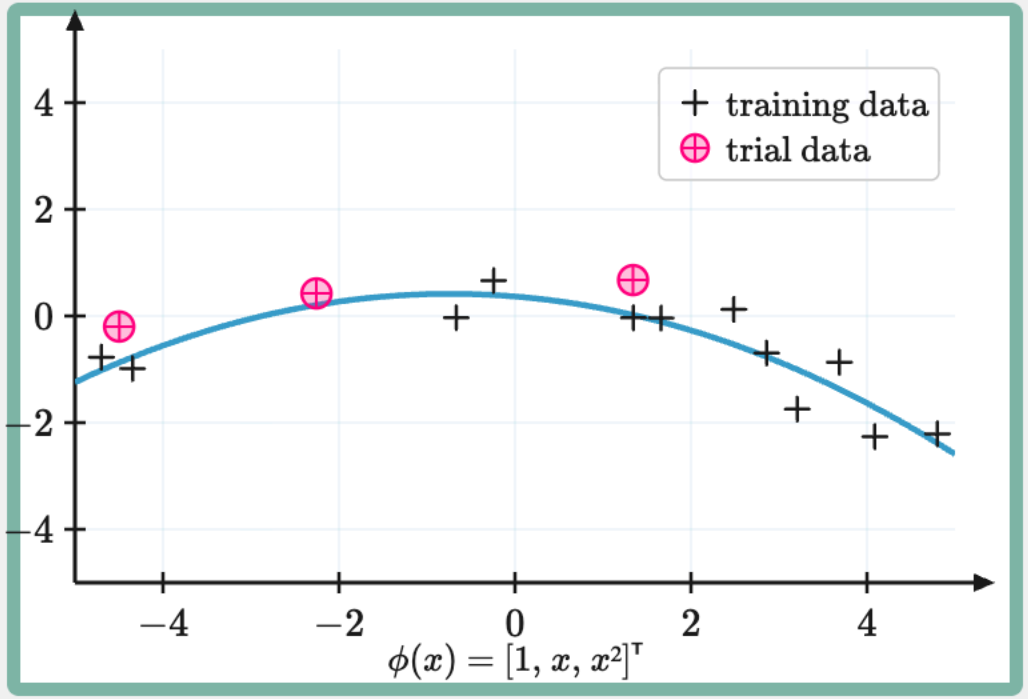
Simple thought experiment

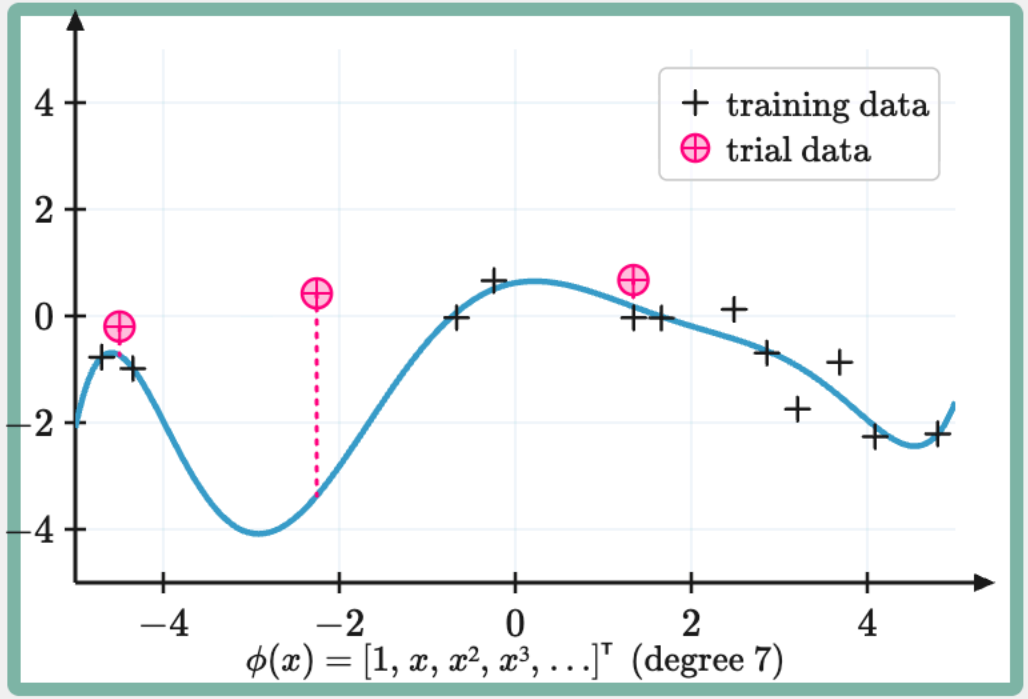




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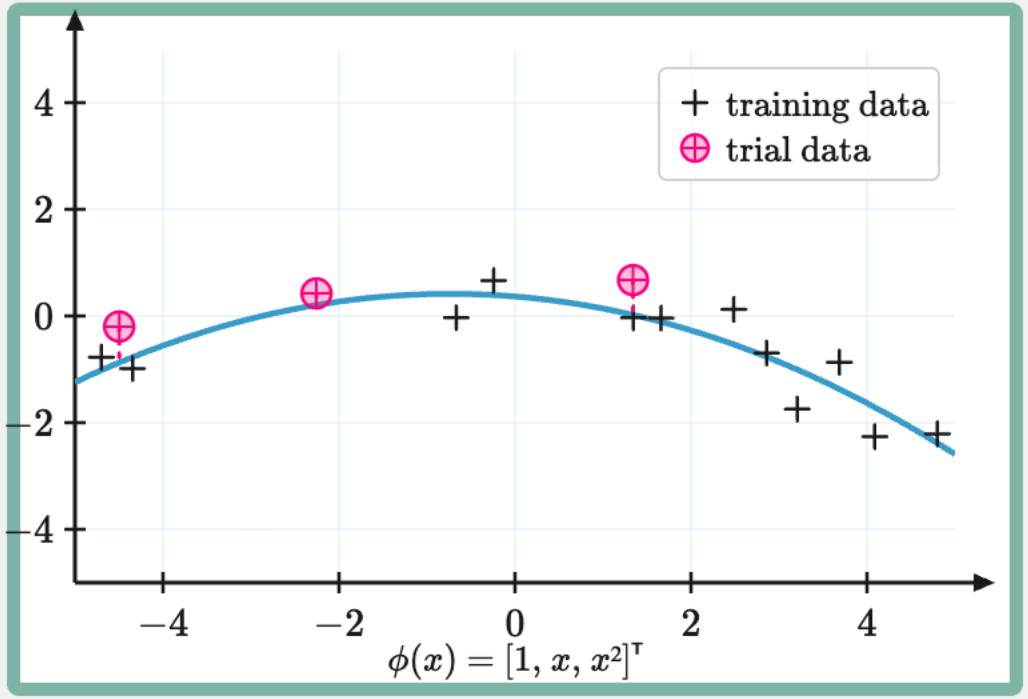
Simple thought experiment



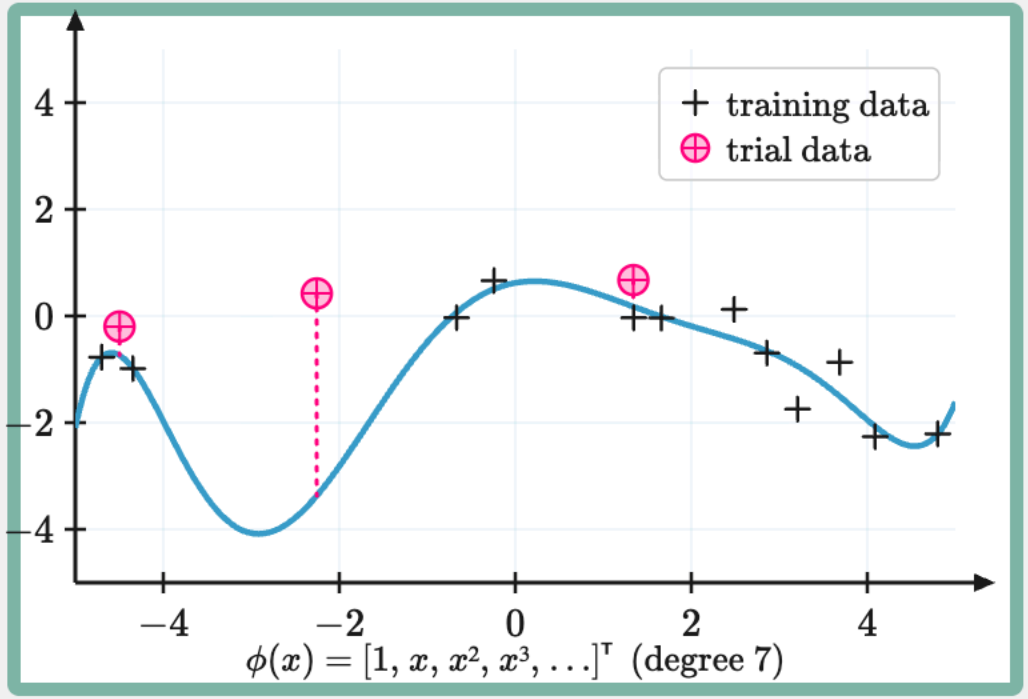


To decide which function is best, let's use both in practice and measure which one performs best

Simple thought experiment



$\hat{R}_k = 0.31 \ \ (k = 3)$ **lower**



$\hat{R}_k = 4.97 \ \ (k = 3)$

To decide which function is best, let's use both in practice and measure which one performs best

Risk function

$$\hat{R}_k(f^\theta) := \underbrace{\frac{1}{k}\sum_{i=1}^{k}}_{\text{averaged over the } k \text{ test points}} \underbrace{\mathcal{L}\left(f^\theta(\boldsymbol{x}^{(i)}), \boldsymbol{y}^{(i)}\right)}_{\text{the loss of the prediction on test point } i}$$

For each input we see in our “trial run” of the models, we measure the “risk” of using our model prediction with a function.

Risk function

$$\hat{R}_k(f^\theta) < \hat{R}_k(g^\omega)$$

We can then say that we prefer to use one model over the other if its risk is lower.

Risk function

$$\frac{1}{k}\sum_{i=1}^{k}\mathcal{L}\left(f^{\theta}(\boldsymbol{x}^{(i)}),\boldsymbol{y}^{(i)}\right) < \frac{1}{k}\sum_{i=1}^{k}\mathcal{L}\left(g^{\omega}(\boldsymbol{x}^{(i)}),\boldsymbol{y}^{(i)}\right)$$

We can then say that we prefer to use one model over the other if its risk is lower.

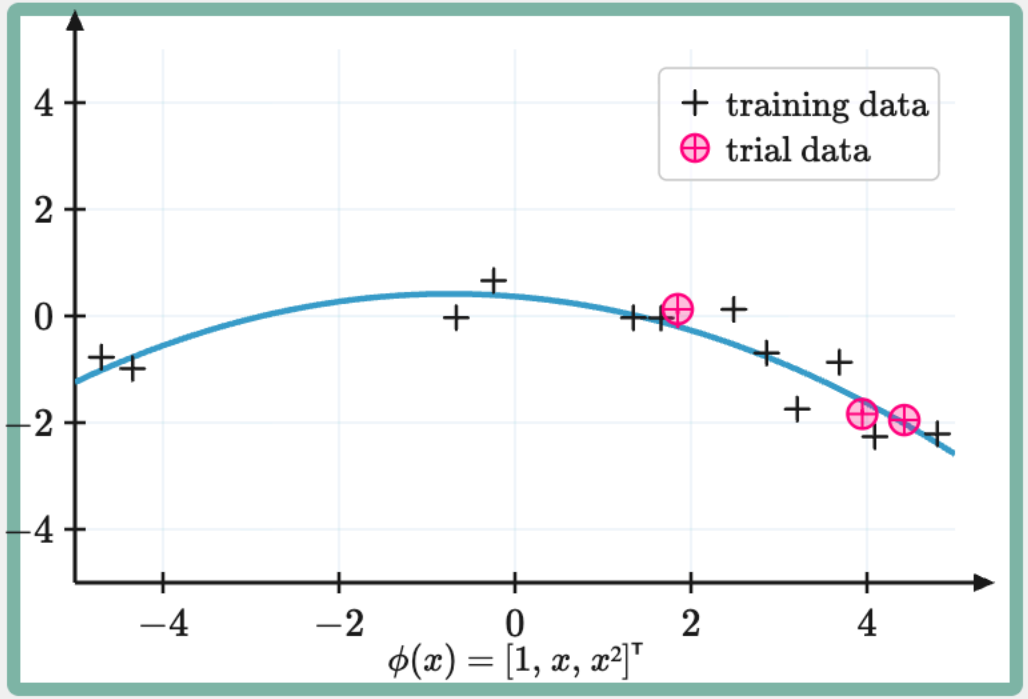
Risk function

$$\frac{1}{k}\sum_{i=1}^{k}\mathcal{L}\left(f^{\theta}(\boldsymbol{x}^{(i)}),\boldsymbol{y}^{(i)}\right) < \frac{1}{k}\sum_{i=1}^{k}\mathcal{L}\left(g^{\omega}(\boldsymbol{x}^{(i)}),\boldsymbol{y}^{(i)}\right)$$

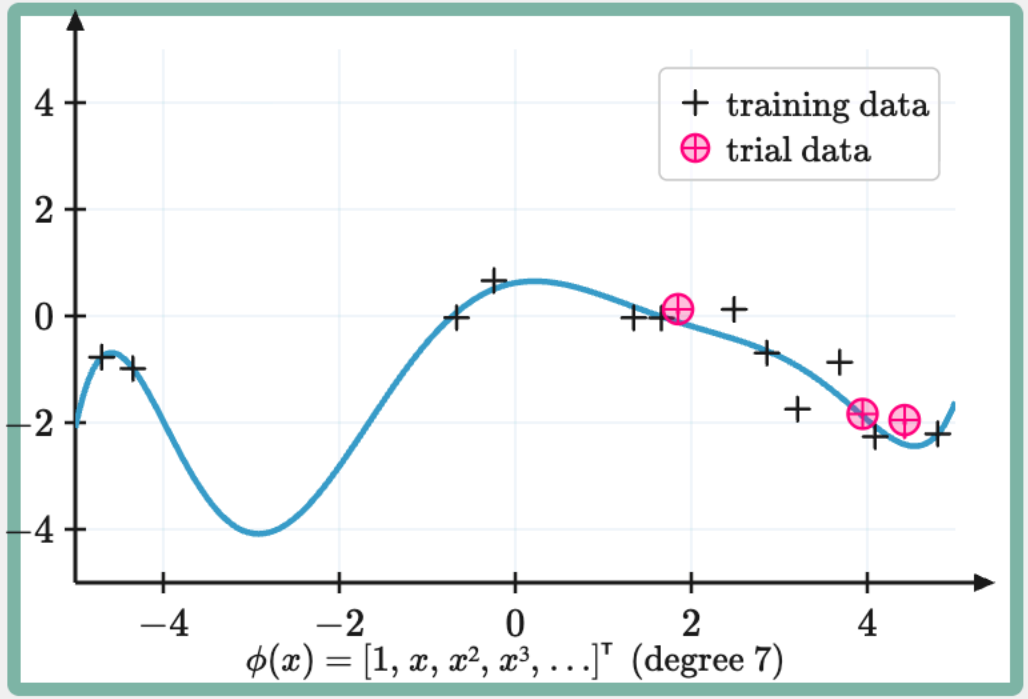
We can then say that we prefer to use one model over the other if its risk is lower.

When do we stop the trial?

When do we stop our trial?



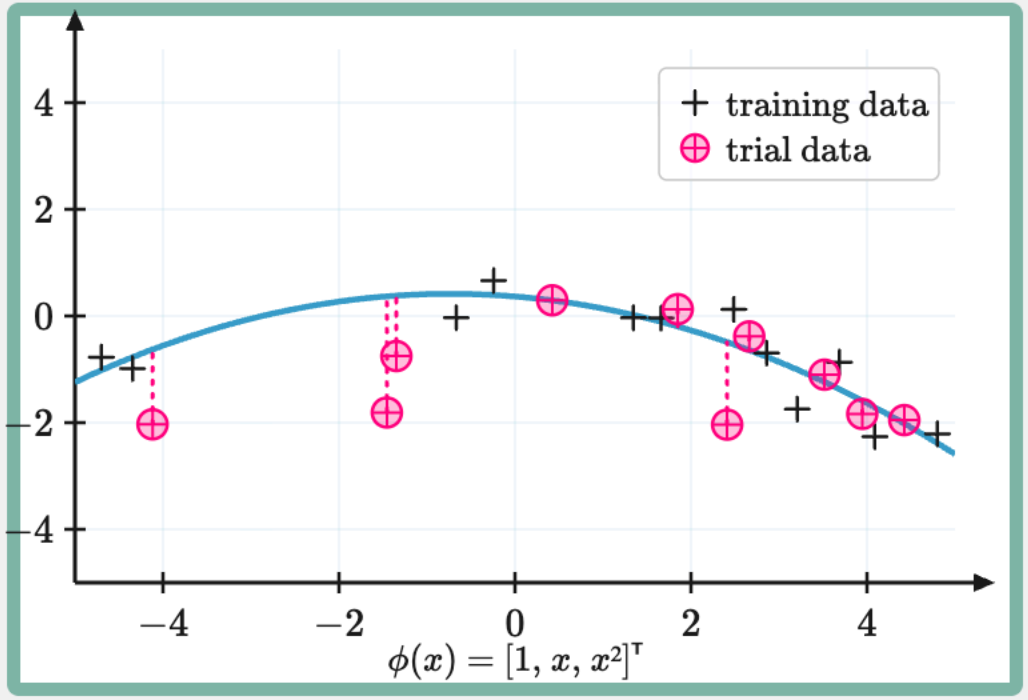
$\hat{R}_k = 0.06 \ \ (k = 3)$ **lower**



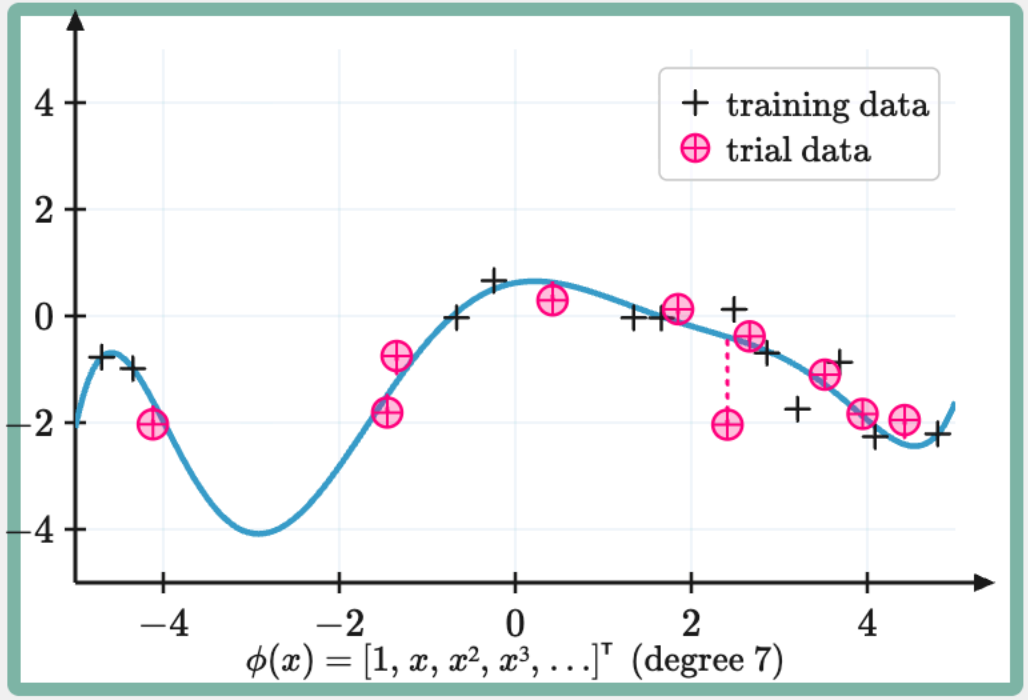
$\hat{R}_k = 0.09 \ \ (k = 3)$

To decide which function is best, let's use both in practice and measure which one performs best (*k*=3)

When do we stop our trial?



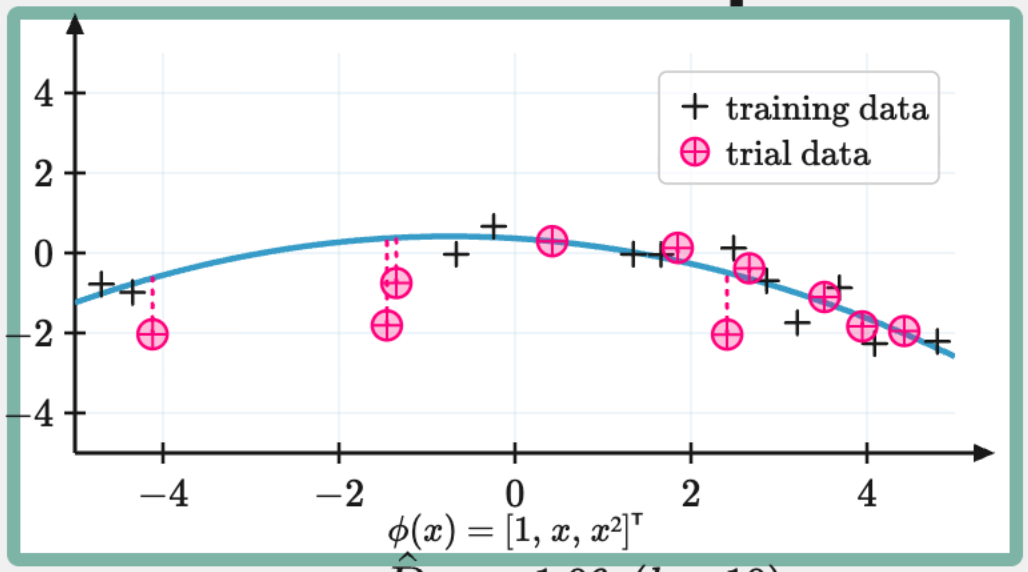
$\hat{R}_k = 1.06 \quad (k = 10)$



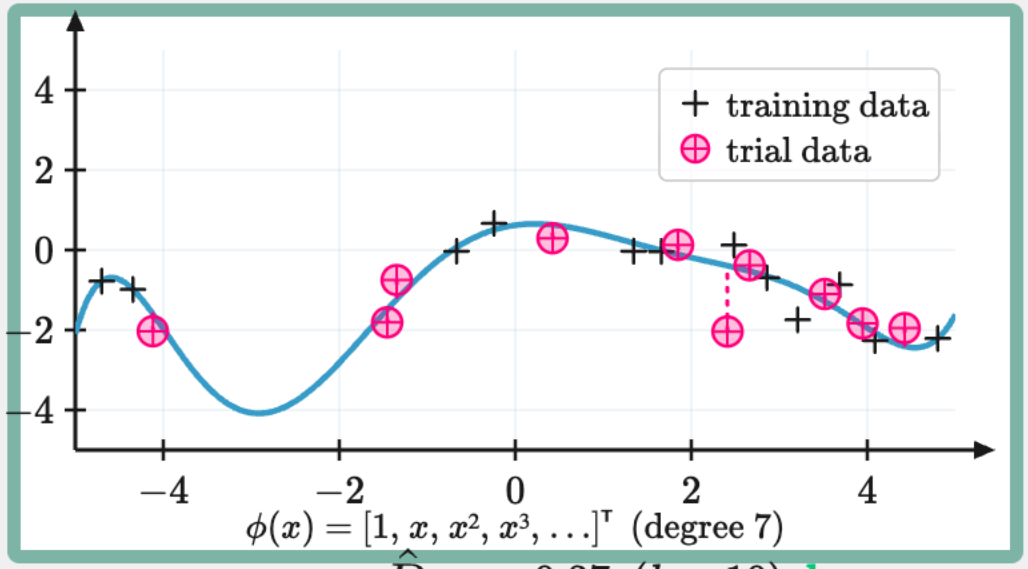
$\hat{R}_k = 0.37 \quad (k = 10)$ **lower**

To decide which function is best, let's use both in practice and measure which one performs best (*k*=10)

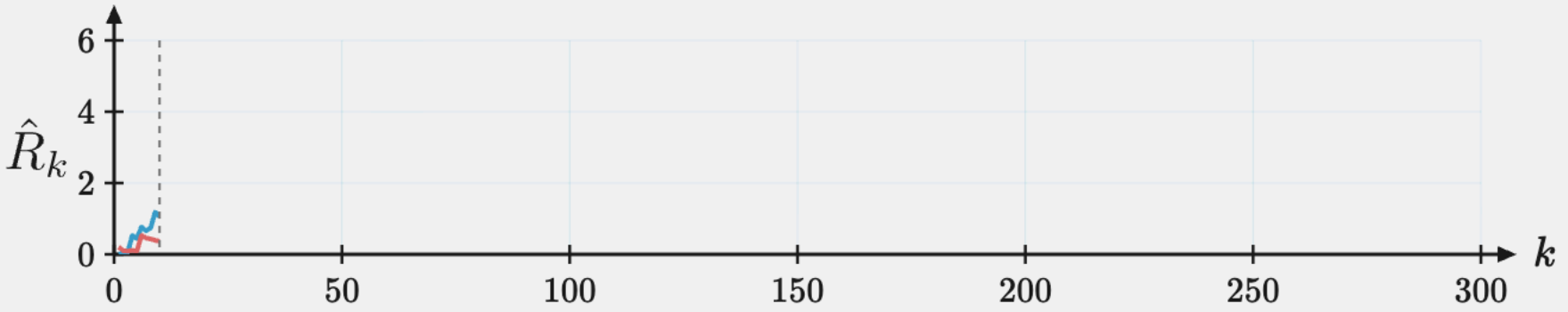
When do we stop our trial?



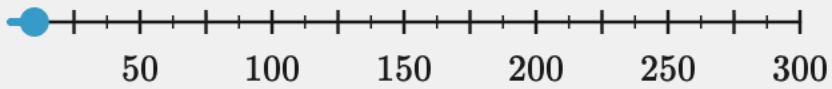
$\hat{R}_k = 1.06 \quad (k = 10)$



$\hat{R}_k = 0.37 \quad (k = 10)$ **lower**



$k = 10$



Modelling things with randomness

Let's consider that our data points are now sampled from a joint distribution:

$$(\boldsymbol{x}^{(i)}, \boldsymbol{y}^{(i)}) \sim P(\mathbf{X}, \mathbf{Y})$$

Each sample is then mapped to a loss value

$$r_i = \mathcal{L}\left(f^{\theta}(\boldsymbol{x}^{(i)}), \boldsymbol{y}^{(i)}\right)$$

And we will describe the distribution of those loss values using a random variable:

$$r_i \sim \mathbf{r}$$

Some qualities of a random variable

Mean or expected value:

$$\mathbb{E}[\mathbf{r}] := \sum_r r\, P(\mathbf{r} = r)$$

Variance:

$$\mathbb{V}[\mathbf{r}] := \mathbb{E}\left[(\mathbf{r} - \mathbb{E}[\mathbf{r}])^2\right] = \sum_r \left(r - \mathbb{E}[\mathbf{r}]\right)^2 P(\mathbf{r} = r)$$

Some qualities of a random variable

$$\mathbb{E}[\mathbf{r}] := \sum_{r} r\, P(\mathbf{r} = r)$$

Expected value is a number

$$\hat{r}_k = \frac{1}{k} \sum_{i=1}^{k} r_i$$

Sample mean is a random value
(described by a random variable)

Difference between estimate and expectation

$$\mathbb{E}[\mathbf{r}] := \sum_{r} r\, P(\mathbf{r} = r)$$

Idealized case: access to all possible sample values, weighted according to exact probability

$$\hat{r}_k = \frac{1}{k} \sum_{i=1}^{k} r_i$$

Practical case: access to “random” samples, no idea of their true probability values

Difference between estimate and expectation

$$\mathbb{E}[\mathbf{r}] := \sum_{r} r\, P(\mathbf{r} = r)$$

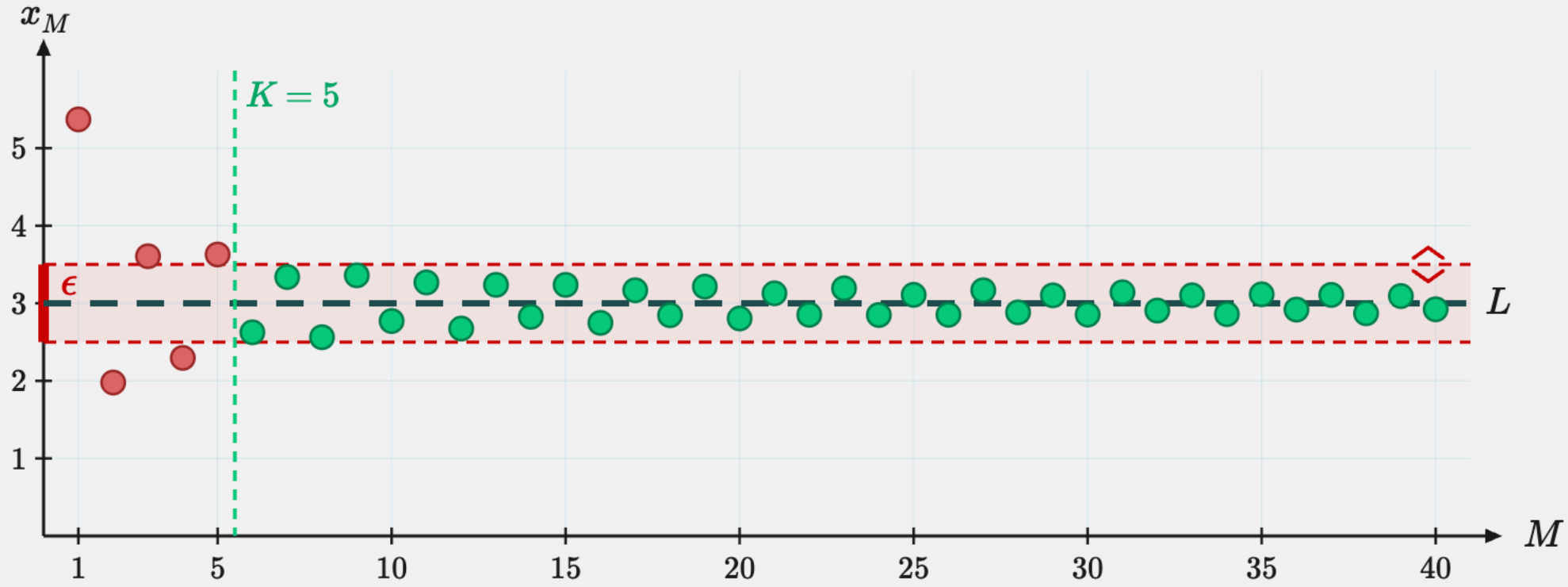
$$\hat{r}_k = \frac{1}{k} \sum_{i=1}^{k} r_i$$

What we want is that in some sense, and for some value '*k*' the following is true:

$$\mathbb{E}[\mathbf{r}] \approx \hat{r}_k$$

Convergence from last lecture

Definition 5.4.1. Convergence (lecture 05) A sequence $x_1, x_2, \ldots$ is said to *converge* to a limit L if for any $\epsilon > 0$ we have an integer K such that $\forall M > K$, $|x_M - L| < \epsilon$.

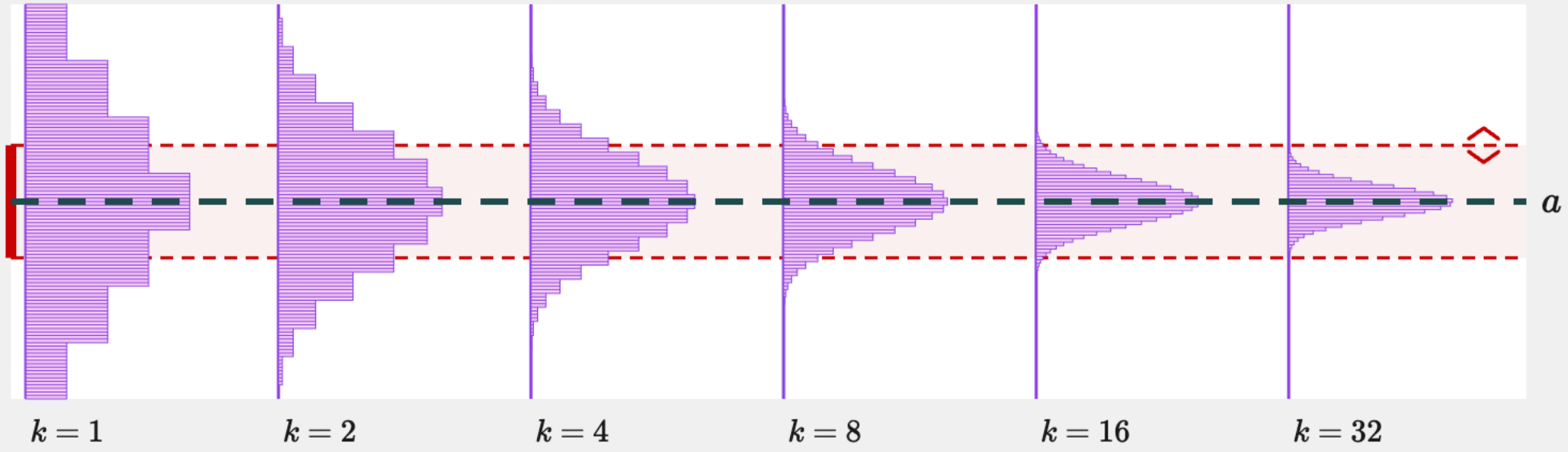


$\epsilon = 0.5$: every x_M with $M > K = 5$ is within ϵ of L: smaller ϵ, later K

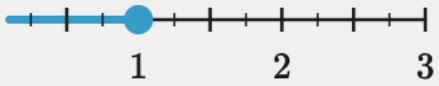
$\epsilon = 0.50$

Convergence in probability

If rather than a series (e.g., arithmetic, geometric) we have a probability distribution, we need a new notion of convergence.



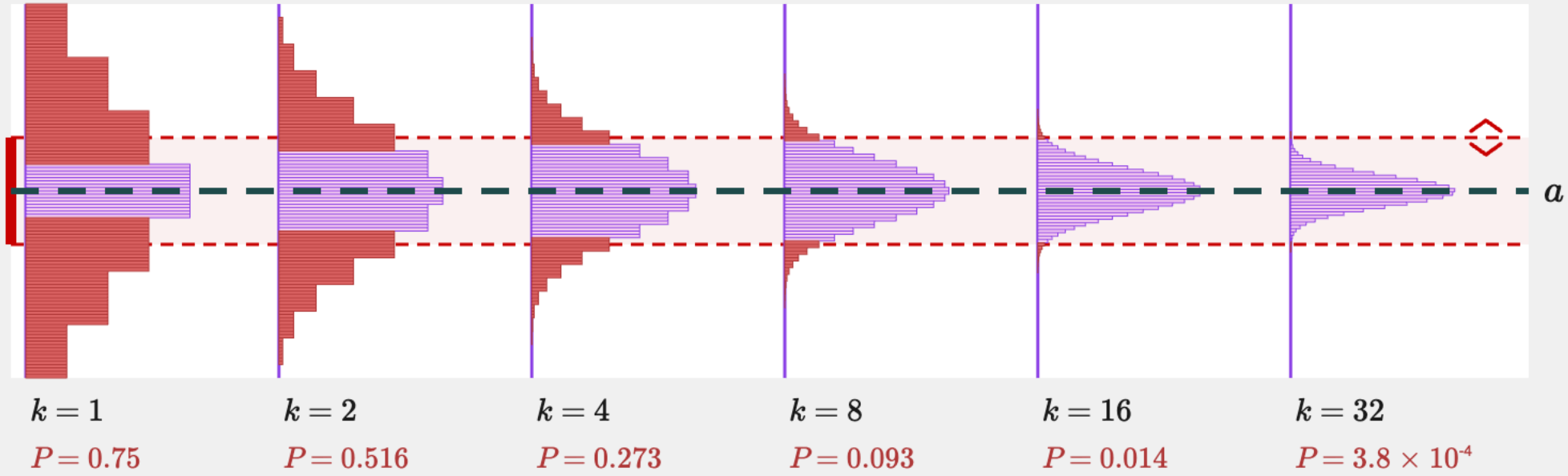
$\epsilon = 1.00$



Convergence in probability

Convergence of a sequence of random variables:

$$\forall \epsilon > 0, \quad \lim_{k \to \infty} P\left(|\hat{r}_k - a| \geq \epsilon\right) = 0$$



$\epsilon = 1.00$

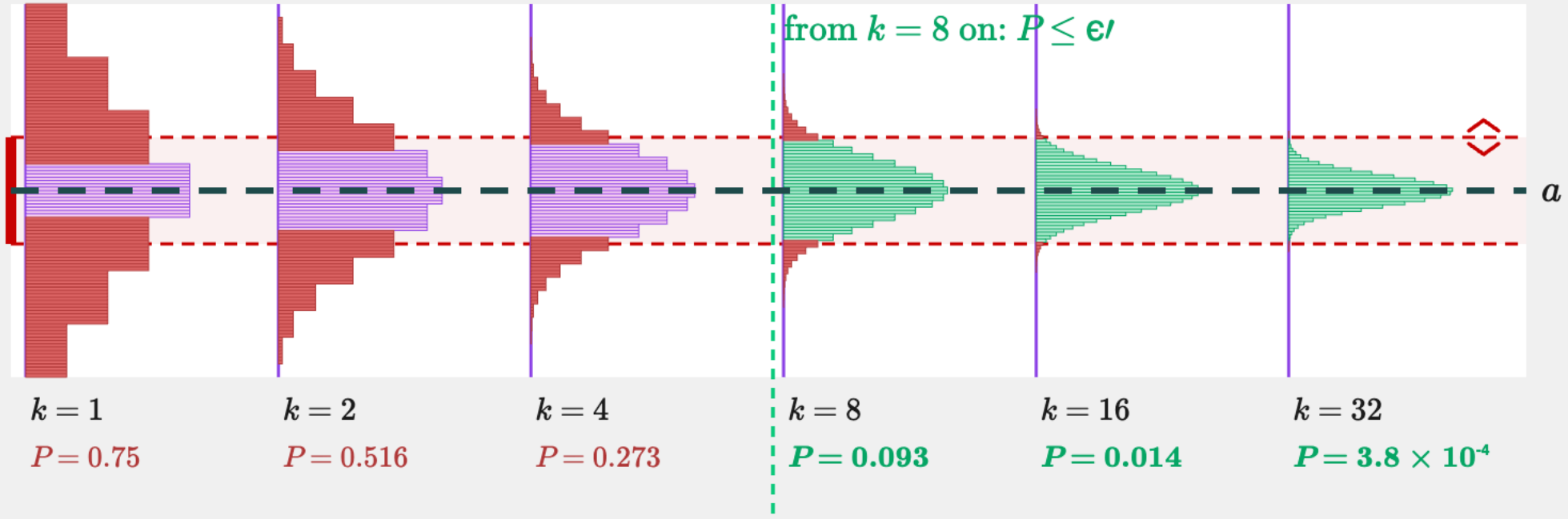
1 2 3

$P(|\hat{r}_k - a| \geq \epsilon)$
red: outside $a \pm \epsilon$

Convergence in probability

Convergence of a sequence of random variables:

$$\forall \epsilon > 0,\ \forall \epsilon' > 0,\ \exists K \text{ s.t. } \forall M > K,\ \ P\left(|\hat{r}_M - a| \geq \epsilon\right) \leq \epsilon'$$



$\epsilon = 1.00$ 1 2 3 $\epsilon' = 0.10$ 0.2 0.4 $P(|\hat{r}_k - a| \geq \epsilon)$

red: outside $a \pm \epsilon$

Does our estimate converge?

$$\hat{r}_k = \frac{1}{k} \sum_{i=1}^{k} r_i$$

Assuming each r_i is i.i.d., do we get convergence in probability to the expectation?

$$\mathbb{E}[\mathbf{r}] \approx \hat{r}_k$$

Does our estimate converge?

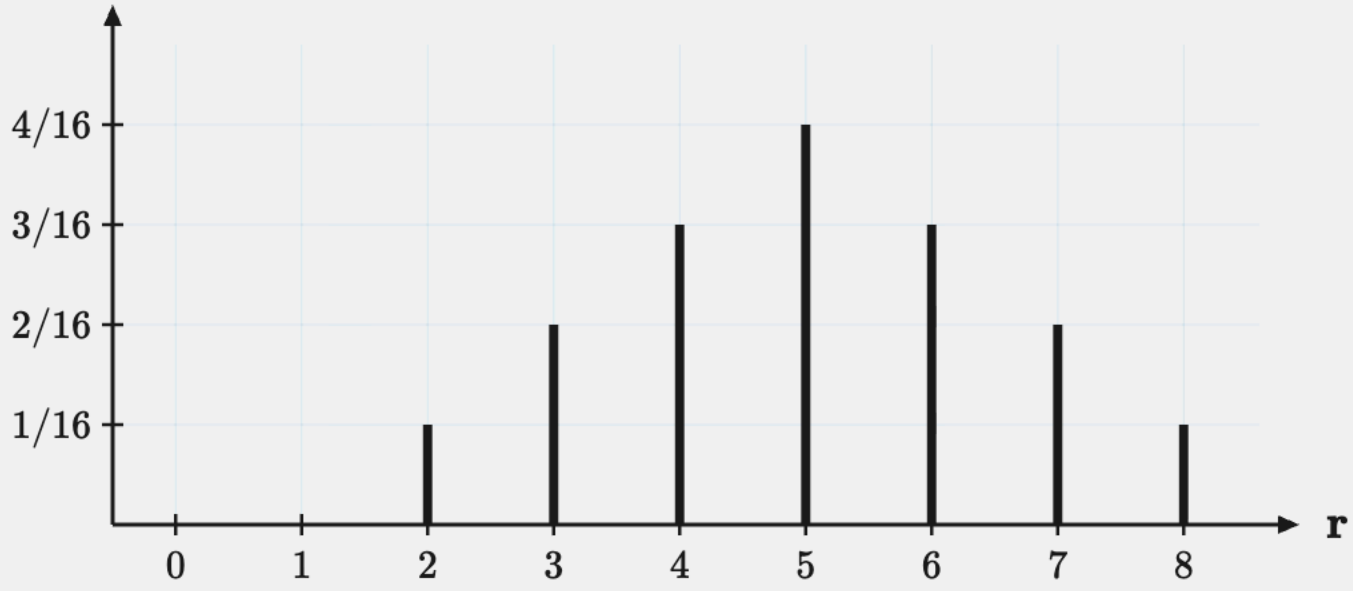
$$\forall \epsilon > 0, \quad \lim_{k \to \infty} P\left(|\hat{r}_k - a| \geq \epsilon\right) = 0$$

Our convergence condition deals in statements of probability, while our mean is a value. We need some way to relate these things.

$$\mathbb{E}[\mathbf{r}] \qquad\qquad P\left(|\hat{r}_k - a| \geq \epsilon\right)$$

Consider the following probability distribution

$$\mathbb{E}[\mathbf{r}] = \sum_{r} r\, P(\mathbf{r} = r)$$



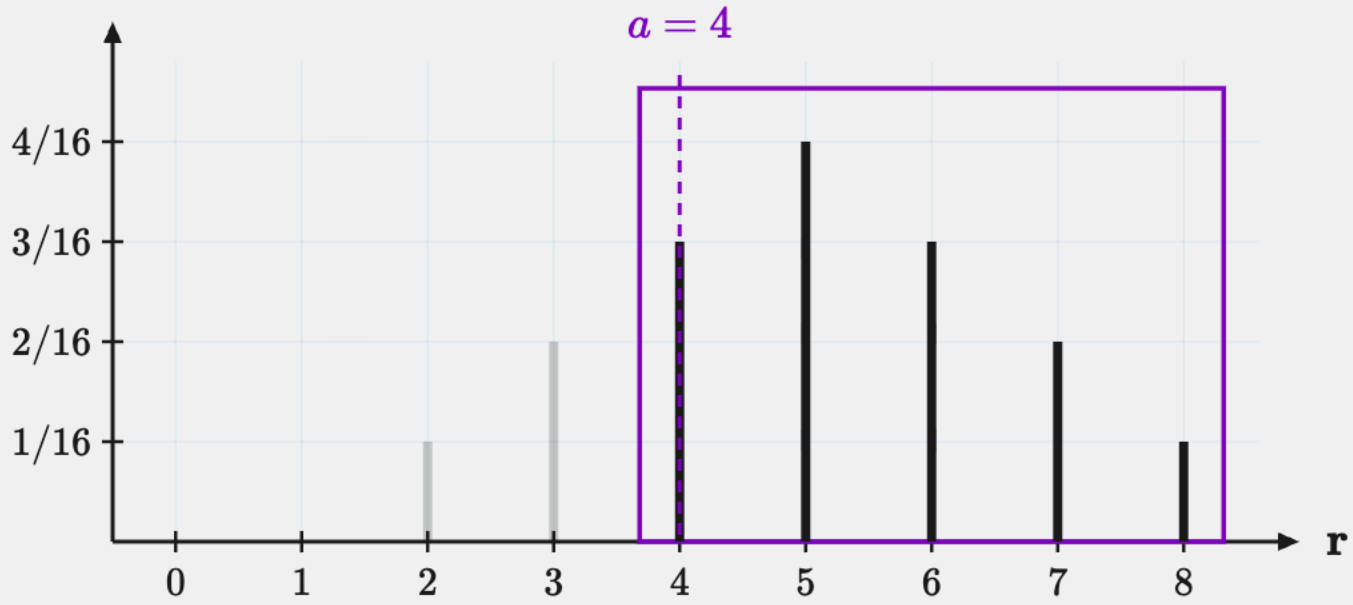
$r\, P(\mathbf{r} = r)$ $0 \cdot 0$ $1 \cdot 0$ $2 \cdot 1/16$ $3 \cdot 2/16$ $4 \cdot 3/16$ $5 \cdot 4/16$ $6 \cdot 3/16$ $7 \cdot 2/16$ $8 \cdot 1/16$

$\mathbb{E}[\mathbf{r}] = 5$



We can begin to bound

$$\sum_{r} r\,P(\mathbf{r}=r) \geq \sum_{r \geq a} r\,P(\mathbf{r}=r)$$



$\mathbb{E}[\mathbf{r}] = 5$

$\sum_{r \geq a} r\,P(\mathbf{r}=r) = 4.5$

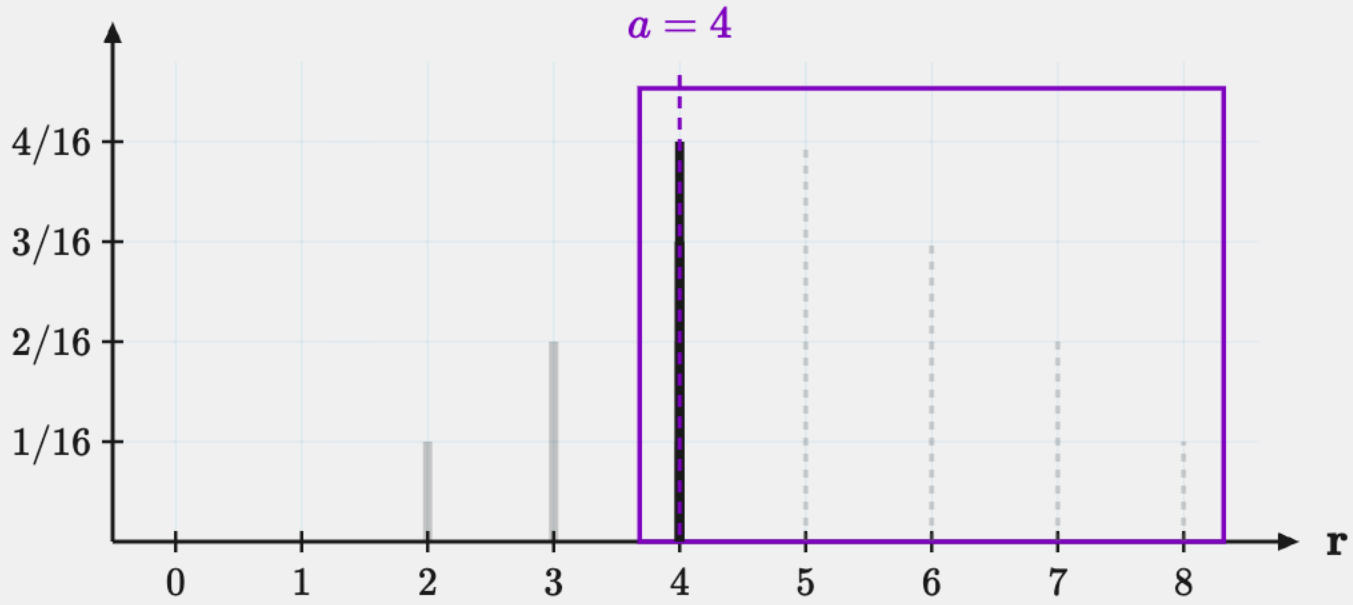
$r\,P(\mathbf{r}=r)$	$0\cdot 0$	$1\cdot 0$	$2\cdot 1/16$	$3\cdot 2/16$	$4\cdot 3/16$	$5\cdot 4/16$	$6\cdot 3/16$	$7\cdot 2/16$	$8\cdot 1/16$
$r\,P(\mathbf{r}=r),\ \ r \geq a$	$0\cdot 0$	$1\cdot 0$	$2\cdot 1/16$	$3\cdot 2/16$	$4\cdot 3/16$	$5\cdot 4/16$	$6\cdot 3/16$	$7\cdot 2/16$	$8\cdot 1/16$

$a = 4.0$



We can begin to bound

$$\sum_{r \geq a} r\, P(\mathbf{r} = r) \geq \sum_{r \geq a} a\, P(\mathbf{r} = r)$$



$\mathbb{E}[\mathbf{r}] = 5$

$\sum_{r \geq a} a\, P(\mathbf{r} = r) = 3.25$

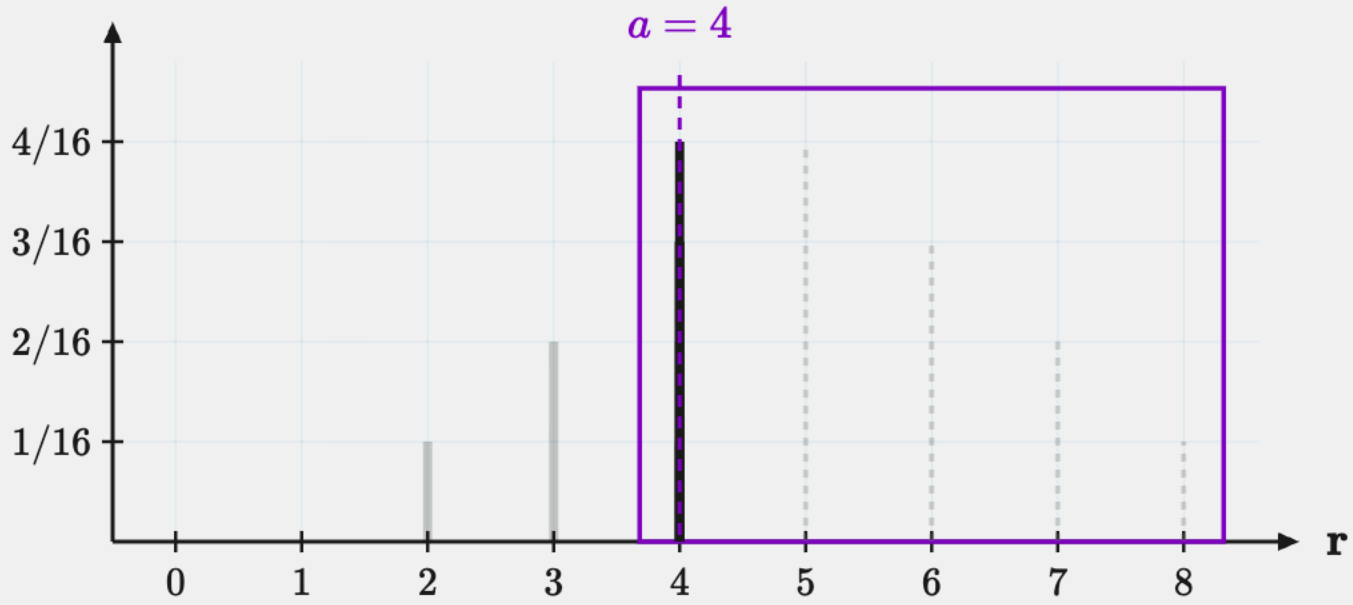
$r\, P(\mathbf{r} = r)$	$0 \cdot 0$	$1 \cdot 0$	$2 \cdot 1/16$	$3 \cdot 2/16$	$4 \cdot 3/16$	$5 \cdot 4/16$	$6 \cdot 3/16$	$7 \cdot 2/16$	$8 \cdot 1/16$
$a\, P(\mathbf{r} = r),\ \ r \geq a$	$0 \cdot 0$	$1 \cdot 0$	$2 \cdot 1/16$	$3 \cdot 2/16$	$4 \cdot 3/16$	$4 \cdot 4/16$	$4 \cdot 3/16$	$4 \cdot 2/16$	$4 \cdot 1/16$

$a = 4.0$



A loose bound on the expectation

$$\sum_{r \geq a} a\, P(\mathbf{r} = r) = a \sum_{r \geq a} P(\mathbf{r} = r) = a\, P(\mathbf{r} \geq a)$$



$\mathbb{E}[\mathbf{r}] = 5$

$a\, P(\mathbf{r} \geq a) = 3.25$

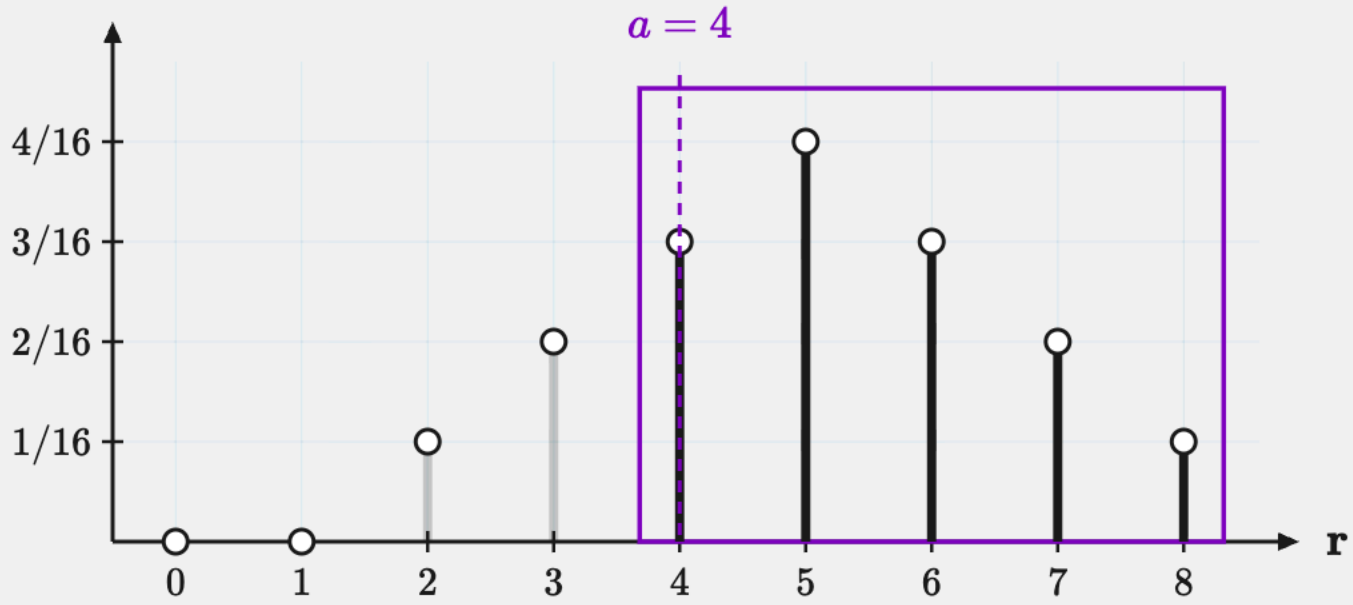
$r\, P(\mathbf{r} = r)$	$0 \cdot 0$	$1 \cdot 0$	$2 \cdot 1/16$	$3 \cdot 2/16$	$4 \cdot 3/16$	$5 \cdot 4/16$	$6 \cdot 3/16$	$7 \cdot 2/16$	$8 \cdot 1/16$
$a\, P(\mathbf{r} = r),\ \ r \geq a$	$0 \cdot 0$	$1 \cdot 0$	$2 \cdot 1/16$	$3 \cdot 2/16$	$4 \cdot 3/16$	$4 \cdot 4/16$	$4 \cdot 3/16$	$4 \cdot 2/16$	$4 \cdot 1/16$

$a = 4.0$



Markov's inequality

$$\mathbb{E}[\mathbf{r}] \geq a\,P(\mathbf{r} \geq a)$$



$\mathbb{E}[\mathbf{r}] = 5$

$a\,P(\mathbf{r} \geq a) = 3.25$

$5 \geq 3.25$

$r\,P(\mathbf{r} = r)$	$0 \cdot 0$	$1 \cdot 0$	$2 \cdot 1/16$	$3 \cdot 2/16$	$4 \cdot 3/16$	$5 \cdot 4/16$	$6 \cdot 3/16$	$7 \cdot 2/16$	$8 \cdot 1/16$
$a\,P(\mathbf{r} = r),\ \ r \geq a$	$0 \cdot 0$	$1 \cdot 0$	$2 \cdot 1/16$	$3 \cdot 2/16$	$4 \cdot 3/16$	$4 \cdot 4/16$	$4 \cdot 3/16$	$4 \cdot 2/16$	$4 \cdot 1/16$

$a = 4.0$



Applying Markov's inequality

Def of variance: $$\mathbb{V}[\mathbf{r}] = \mathbb{E}\Big[\big(\mathbb{E}[\mathbf{r}] - \mathbf{r}\big)^2\Big]$$

Markov's Inequality: $$\mathbb{E}[\mathbf{r}] \geq a\, P(\mathbf{r} \geq a)$$

Combination:

Applying Markov's inequality

Def of variance: $$\mathbb{V}[\mathbf{r}] = \mathbb{E}\Big[\big(\mathbb{E}[\mathbf{r}] - \mathbf{r}\big)^2\Big]$$

Markov's Inequality: $$\mathbb{E}[\mathbf{r}] \geq a\, P(\mathbf{r} \geq a)$$

Combination: $$\mathbb{E}\Big[\big(\mathbb{E}[\mathbf{r}] - \mathbf{r}\big)^2\Big] \geq a^2\, P\Big(\big(\mathbb{E}[\mathbf{r}] - \mathbf{r}\big)^2 \geq a^2\Big)$$

Markov for the random variable $(\mathbb{E}[\mathbf{r}] - \mathbf{r})^2 \geq 0$ (orange), with a^2 in place of a (purple)

Applying Markov's inequality

Def of variance: $$\mathbb{V}[\mathbf{r}] = \mathbb{E}\Big[\big(\mathbb{E}[\mathbf{r}] - \mathbf{r}\big)^2\Big]$$

Markov's Inequality: $$\mathbb{E}[\mathbf{r}] \geq a\, P(\mathbf{r} \geq a)$$

Combination: $$\mathbb{E}\Big[\big(\mathbb{E}[\mathbf{r}] - \mathbf{r}\big)^2\Big] \geq a^2\, P\Big(\big(\mathbb{E}[\mathbf{r}] - \mathbf{r}\big)^2 \geq a^2\Big)$$

$$= a^2\, P\big(|\mathbb{E}[\mathbf{r}] - \mathbf{r}| \geq a\big)$$

both sides are non-negative: $(\mathbb{E}[\mathbf{r}] - \mathbf{r})^2 \geq a^2$ exactly when $|\mathbb{E}[\mathbf{r}] - \mathbf{r}| \geq a$

Markov's inequality applied to the variance

Recalling $\mathbb{V}[\mathbf{r}] := \sigma^2$

$$\sigma^2 \geq a^2 \, P\big(|\mathbb{E}[\mathbf{r}] - \mathbf{r}| \geq a\big)$$

the left-hand side of the last line is the variance, σ^2

Markov's inequality applied to the variance

Recalling $\mathbb{V}[\mathbf{r}] := \sigma^2$

$$\sigma^2 \geq a^2 \, P\big(|\mathbb{E}[\mathbf{r}] - \mathbf{r}| \geq a\big)$$

$$\frac{\sigma^2}{a^2} \geq P\big(|\mathbb{E}[\mathbf{r}] - \mathbf{r}| \geq a\big)$$

divide both sides by $a^2 > 0$

Chebyshev's inequality

Recalling $\mathbb{V}[\mathbf{r}] := \sigma^2$

$$\sigma^2 \geq a^2 \, P\big(|\mathbb{E}[\mathbf{r}] - \mathbf{r}| \geq a\big)$$

$$\frac{\sigma^2}{a^2} \geq P\big(|\mathbb{E}[\mathbf{r}] - \mathbf{r}| \geq a\big)$$

Letting $a = c\,\sigma$:

$$\frac{1}{c^2} \geq P\big(|\mathbb{E}[\mathbf{r}] - \mathbf{r}| \geq c\,\sigma\big)$$

$\sigma^2/(c\sigma)^2 = 1/c^2$

Applying Markov's inequality

Markov's Inequality:

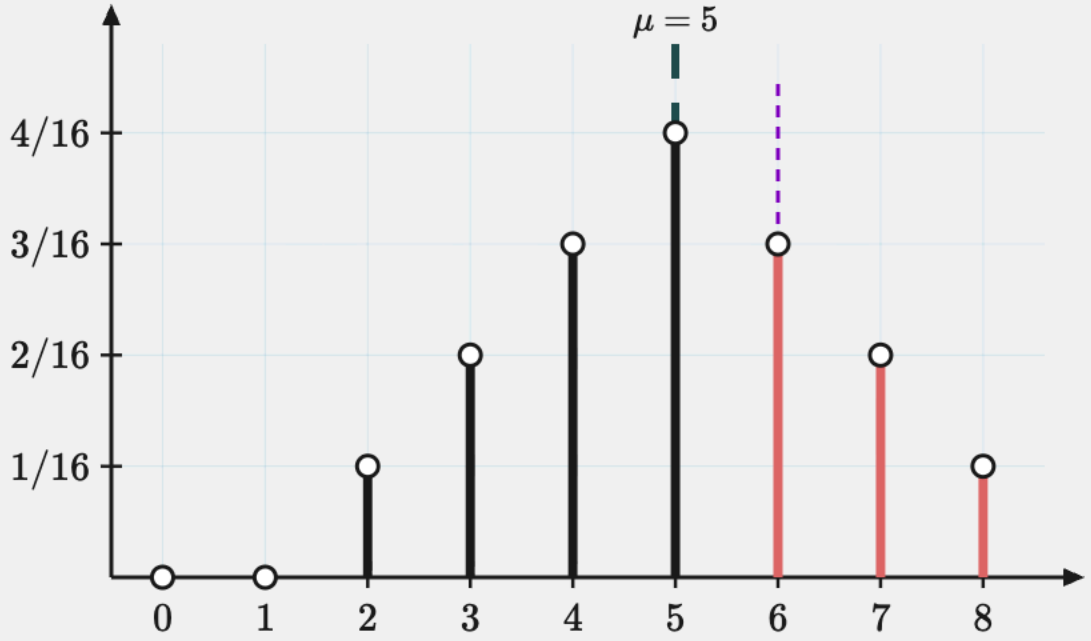
$$\mathbb{E}[\mathbf{r}] \geq a\, P(\mathbf{r} \geq a)$$

Chebyshev's Inequality:

$$\frac{1}{c^2} \geq P\big(|\mathbb{E}[\mathbf{r}] - \mathbf{r}| \geq c\,\sigma\big)$$

Markov and Chebyshev, live

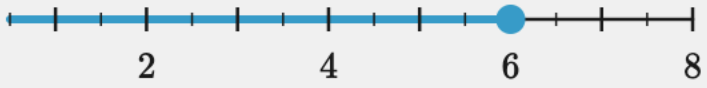
Markov



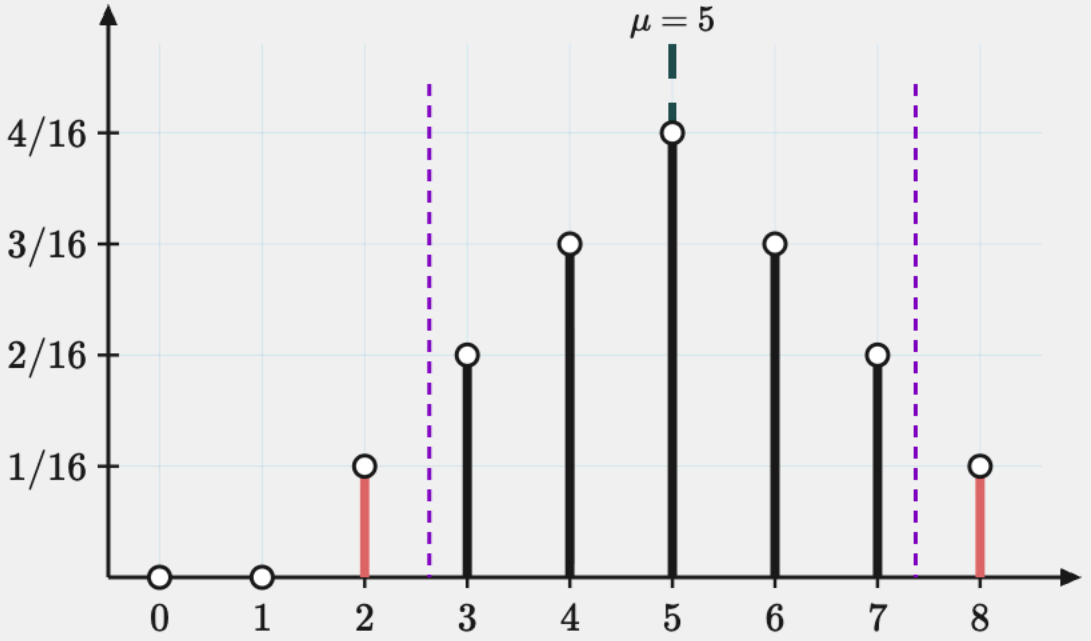
exact: $P(\mathbf{r} \geq 6) = 0.375$

Markov: $\leq \mathbb{E}[\mathbf{r}]/a = 0.833$

$a = 6.00$



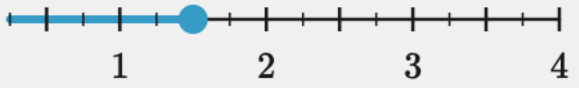
Chebyshev



exact: $P(|\mathbf{r} - \mu| \geq 2.37) = 0.125$

Chebyshev: $\leq \sigma^2/a^2 = 0.444 \;\; (\sigma^2 = 2.5)$

$a \,/\, \sigma = 1.50$



Revisiting the mean of our estimator

$$\hat{r}_k = \frac{r_1 + r_2 + \cdots + r_k}{k}$$

$$\mathbb{E}[\hat{r}_k] = \frac{\mathbb{E}[r_1] + \mathbb{E}[r_2] + \cdots + \mathbb{E}[r_k]}{k}$$

Revisiting our estimator

$$\hat{r}_k = \frac{r_1 + r_2 + \cdots + r_k}{k}$$

$$\mathbb{E}[\hat{r}_k] = \frac{\mathbb{E}[r_1] + \mathbb{E}[r_2] + \cdots + \mathbb{E}[r_k]}{k}$$

$$\mathbb{E}[\hat{r}_k] = \frac{\mu_{\mathbf{r}} + \mu_{\mathbf{r}} + \cdots + \mu_{\mathbf{r}}}{k} = \frac{k\,\mu_{\mathbf{r}}}{k} = \mu_{\mathbf{r}}$$

identically distributed: every $\mathbb{E}[r_i]$ is the same, $\mu_{\mathbf{r}} = \mathbb{E}[\mathbf{r}]$
k equal terms

This seems like what we wanted? What is missing?

What is missing?

Convergence of a sequence of random variables:

$$\forall \epsilon > 0, \quad \lim_{k \to \infty} P\big(|\hat{r}_k - a| \geq \epsilon\big) = 0$$

$$\mathbb{E}[\hat{r}_k] = \mu_{\mathbf{r}}$$

We also need that the variance shrinks in a particular way!

Consider how the variance changes

$$\hat{r}_k = \frac{r_1 + r_2 + \cdots + r_k}{k}$$

$$\mathbb{V}[\hat{r}_k] = \mathbb{V}\left[\frac{r_1 + r_2 + \cdots + r_k}{k}\right]$$ the variance of the sample mean

$$= \frac{\mathbb{V}[r_1] + \mathbb{V}[r_2] + \cdots + \mathbb{V}[r_k]}{k^2}$$

$\mathbb{V}[X/k] = \mathbb{V}[X]/k^2$, and for **independent** draws the variance of a sum is the sum of the variances

Consider how the variance changes

$$\mathbb{V}[\hat{r}_k] = \mathbb{V}\left[\frac{r_1 + r_2 + \cdots + r_k}{k}\right]$$

$$= \frac{\mathbb{V}[r_1] + \mathbb{V}[r_2] + \cdots + \mathbb{V}[r_k]}{k^2}$$

Take a second to simplify this and compute what the variance is in terms of σ

Consider how the variance changes

$$= \frac{\mathbb{V}[r_1] + \mathbb{V}[r_2] + \cdots + \mathbb{V}[r_k]}{k^2}$$

$$= \frac{\sigma^2 + \sigma^2 + \cdots + \sigma^2}{k^2}$$

each draw has variance σ^2

$$= \frac{k\,\sigma^2}{k^2} = \frac{\sigma^2}{k}$$

k equal terms

it shrinks like 1/*k*

Relating this to our convergence definition

$$= \frac{\mathbb{V}[r_1] + \mathbb{V}[r_2] + \cdots + \mathbb{V}[r_k]}{k^2}$$

$$= \frac{\sigma^2 + \sigma^2 + \cdots + \sigma^2}{k^2}$$

each draw has variance σ^2

$$= \frac{k\,\sigma^2}{k^2} = \frac{\sigma^2}{k}$$

k equal terms

it shrinks like 1/*k*

This looks like something we have seen before!

Chebyshev for our estimator

We have shown that our estimate $\hat{r}_k$ has mean μ and variance σ^2/k:

$$\mathbb{E}[\hat{r}_k] = \mu \qquad \mathbb{V}[\hat{r}_k] = \frac{\sigma^2}{k}$$

Recalling Chebyshev's inequality:

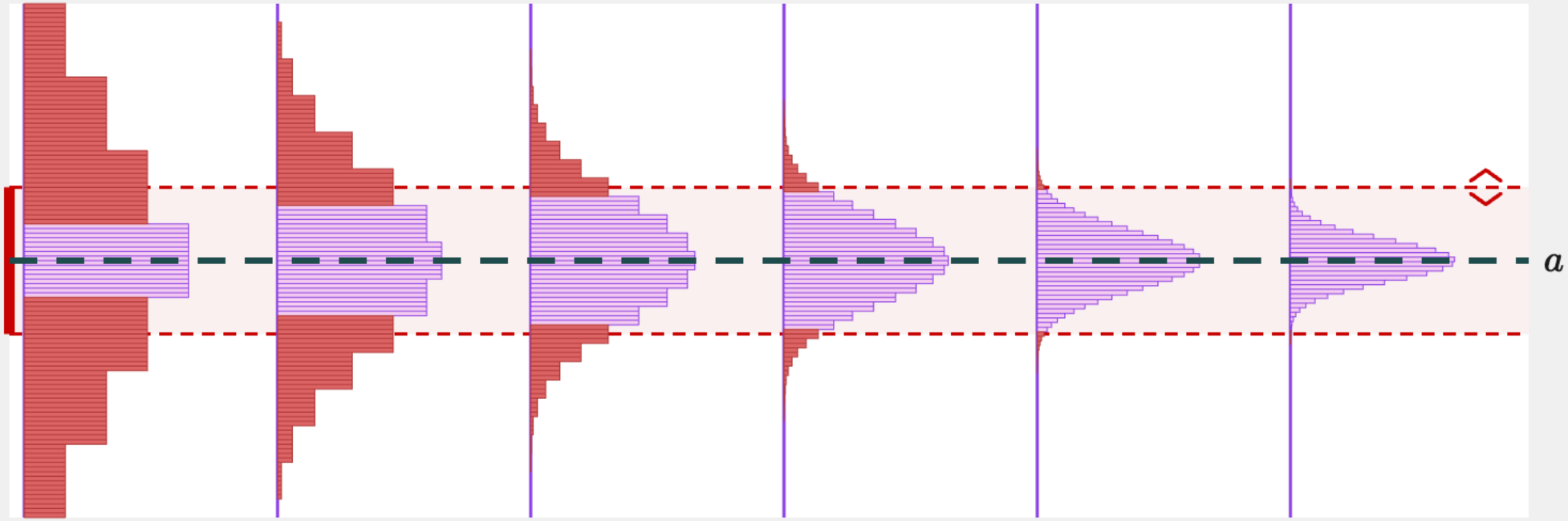
$$P\left(|\mu - \hat{r}_k| \geq \epsilon\right) \leq \frac{\sigma^2}{k\,\epsilon^2}$$

writing ϵ for *a*, as is conventional

Weak law of large numbers

$$P\left(|\mu - \hat{r}_k| \geq \epsilon\right) \leq \frac{\sigma^2}{k\,\epsilon^2}$$

The weak law, live



$k = 1$	$k = 2$	$k = 4$	$k = 8$	$k = 16$	$k = 32$
$P = 0.75$	$P = 0.516$	$P = 0.273$	$P = 0.093$	$P = 0.014$	$P = 3.8 \times 10^{-4}$
≤ 1	≤ 1	≤ 0.625	≤ 0.313	≤ 0.156	≤ 0.078

$\epsilon = 1.00$

1 2 3

$\sigma^2 = 2.5$

Chebyshev: $P \leq \sigma^2/(k\epsilon^2)$

Answering our original question

How long should our trial run be?

$$P\left(|\mu - \hat{r}_k| \geq \epsilon\right) \leq \frac{\sigma^2}{k\,\epsilon^2}$$

Answering our original question

How long should our trial run be?

$$P\left(|\mu - \hat{r}_k| \geq \epsilon\right) \leq \frac{\sigma^2}{k\,\epsilon^2}$$

First we can write the above in a more general form:

$$P\left(|\hat{r}_k - \mu| \geq \epsilon\right) \leq \delta$$

Probably approximately correct

As a brief aside, this is a standard "statistical guarantee"

$$P\left(|\hat{r}_k - \mu| \geq \epsilon\right) \leq \delta$$

Which you may often see written in a "probably approximately correct" form:

$$P\Big(\underbrace{|\hat{r}_k - \mu| \leq \epsilon}_{\text{approximately correct}}\Big) \geq \underbrace{1 - \delta}_{\text{probably}}$$

Answering our original question

How long should our trial run be?

$$P\left(|\mu - \hat{r}_k| \geq \epsilon\right) \leq \frac{\sigma^2}{k\,\epsilon^2}$$

If we know the guarantee we want to get, the idea here is to solve for '*k*' in terms of δ and ϵ

$$P\left(|\hat{r}_k - \mu| \geq \epsilon\right) \leq \delta$$

Answering our original question

How long should our trial run be?

$$P\left(|\mu - \hat{r}_k| \geq \epsilon\right) \leq \frac{\sigma^2}{k\,\epsilon^2}$$

If we know the guarantee we want to get, the idea here is to solve for '*k*' in terms of δ and ϵ

$$\frac{\sigma^2}{k\,\epsilon^2} \leq \delta$$

enough: the bound of the weak law at most δ

Answering our original question

How long should our trial run be?

$$P\left(|\mu - \hat{r}_k| \geq \epsilon\right) \leq \frac{\sigma^2}{k\,\epsilon^2}$$

If we know the guarantee we want to get, the idea here is to solve for '*k*' in terms of δ and ϵ

$$\frac{\sigma^2}{k\,\epsilon^2} \leq \delta$$

enough: the bound of the weak law at most δ

Answering our original question

$$\frac{\sigma^2}{k\,\epsilon^2} \leq \delta$$

$$\frac{\sigma^2}{k\,\epsilon^2} \leq \delta \Longrightarrow \sigma^2 \leq \delta\, k\, \epsilon^2$$

multiply both sides by $k\epsilon^2 > 0$

Answering our original question

$$\frac{\sigma^2}{k\,\epsilon^2} \leq \delta$$

$$\frac{\sigma^2}{k\,\epsilon^2} \leq \delta \implies \sigma^2 \leq \delta\, k\, \epsilon^2$$

$$\implies k \geq \frac{\sigma^2}{\delta\,\epsilon^2}$$

divide both sides by $\delta\epsilon^2 > 0$

Answering our original question

So if '*k*' is bigger than this quantity, then we know we have satisfied the statistical guarantee!

$$k \geq \frac{\sigma^2}{\delta\,\epsilon^2}$$

A distribution free answer?

So if '*k*' is bigger than this quantity, then we know we have satisfied the statistical guarantee!

$$k \geq \frac{\sigma^2}{\delta\,\epsilon^2}$$

Because we do not know sigma, this is a distribution dependent bound.

A distribution free answer!

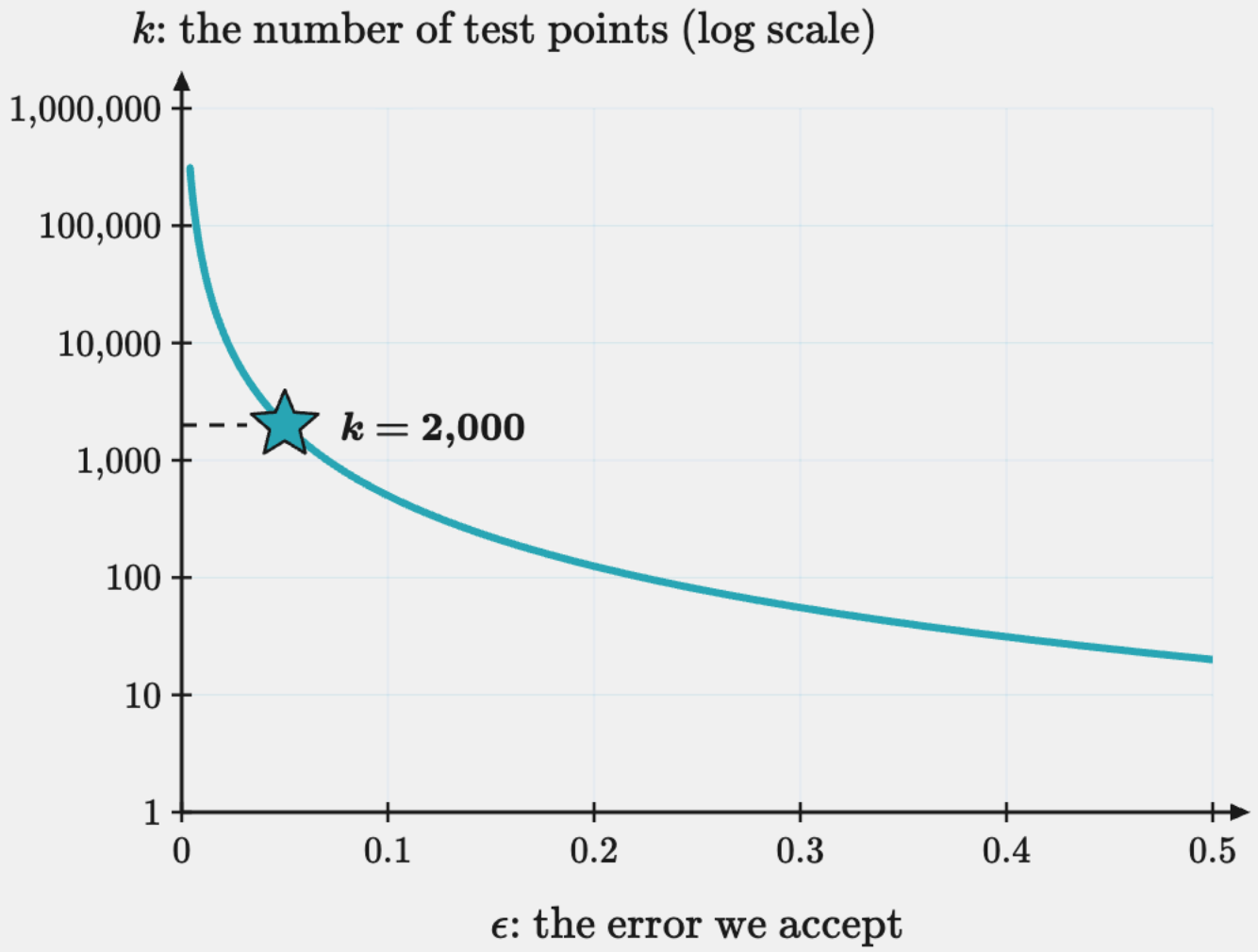
So if '*k*' is bigger than this quantity, then we know we have satisfied the statistical guarantee!

$$k \geq \frac{1/4}{\delta\,\epsilon^2} = \frac{1}{4\,\delta\,\epsilon^2}$$

$$\mathbf{r} \in [0,1] \implies \sigma^2 \leq \tfrac{1}{4}$$

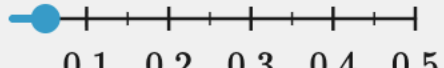
Because we do not know sigma, this is a distribution dependent bound. If we know something about the random variable of interest e.g., that it is bounded between 0 and 1, then we can say more.

How long should the trial be?

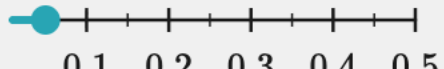


$$k \geq \frac{1}{4\,\delta\,\epsilon^2}$$

$\epsilon = 0.05$



$\delta = 0.05$



2,000 test points are enough:

$P(|\mu - \hat{r}_k| \geq 0.05) \leq \mathbf{0.05}$